Introduction to the Theory of Plate Tectonics

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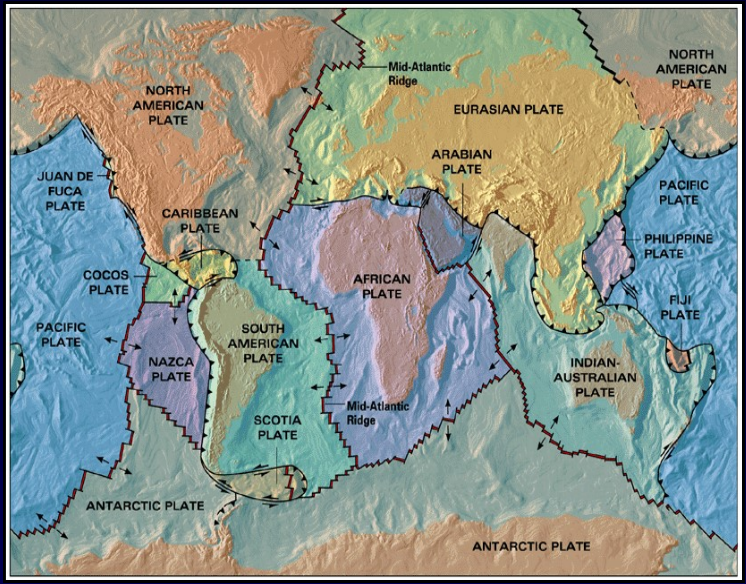
L16: 12 Feb. 2024
L19: 19 Feb. 2024

Plate ?



Outer rigid layer of the earth is divided into number of zones. These zones are called Plates / Lithospheric Plates / Tectonic Plates.

Tectonics ?

Tectonics is a Greek word signifying “to build”.

Plate Tectonics ?

Plate tectonics is the study of forces operating within our planet Earth that give rise to continents, ocean basins, mountain ranges, earthquake belts………

It provides comprehensive explanation about how are continents (i) breaking apart, (ii) drifting from place to place, (iii) colliding, and (iv) grinding against each other.

It explains (i) rising up of terrestrial mountain ranges, (ii) opening and closing of oceans, (iii) occurrences of violent earthquakes and fiery volcanoes and many more ……

Theory of Plate Tectonics

Combination of Two Ideas

Continental Drift

Sea-Floor Spreading

- Evolution of theory of Plate Tectonics
- Basic Concepts of Plate Tectonics
- Plate Movements and Boundaries

Evolution of the Theory of Plate Tectonics

Most Fundamental Questions → But No Answer

- Was the Earth cooling, heating or staying at a constant temperature ?
- Was the Earth contracting, expanding or retaining its size ?
- Did the interior of the Earth convect ?
- Did continents drift ?
- Why were certain parts of the Earth more affected by earthquakes and volcanoes ?
- Why did large mountainous regions occur on margins of continents ?

Evolution of the Theory of Plate Tectonics

No one knew for sure. Nor, there was any tools & techniques in the past to know.

- Many hypotheses were forwarded to answer those questions.
- None answered all the questions at all localities.
- However, theory of continental drift was welcomed by some researchers initially.

Continental Drift

(First step in development of plate tectonics)

- The theory was proposed by Alfred Wegener, who was a meteorologist (geophysicist) from Germany.
- The theory of continental drift basically explains gradual drifting of continents from Pangaea to their present-day positions.
- Alfred Wegener first published this theory in 1912.

Though , Frank Bursley Taylor (1860-1938) and Howard Baker were early proponents, the theory is associated most closely with Alfred Wegener (1880-1930).

Theory of Continental Drift

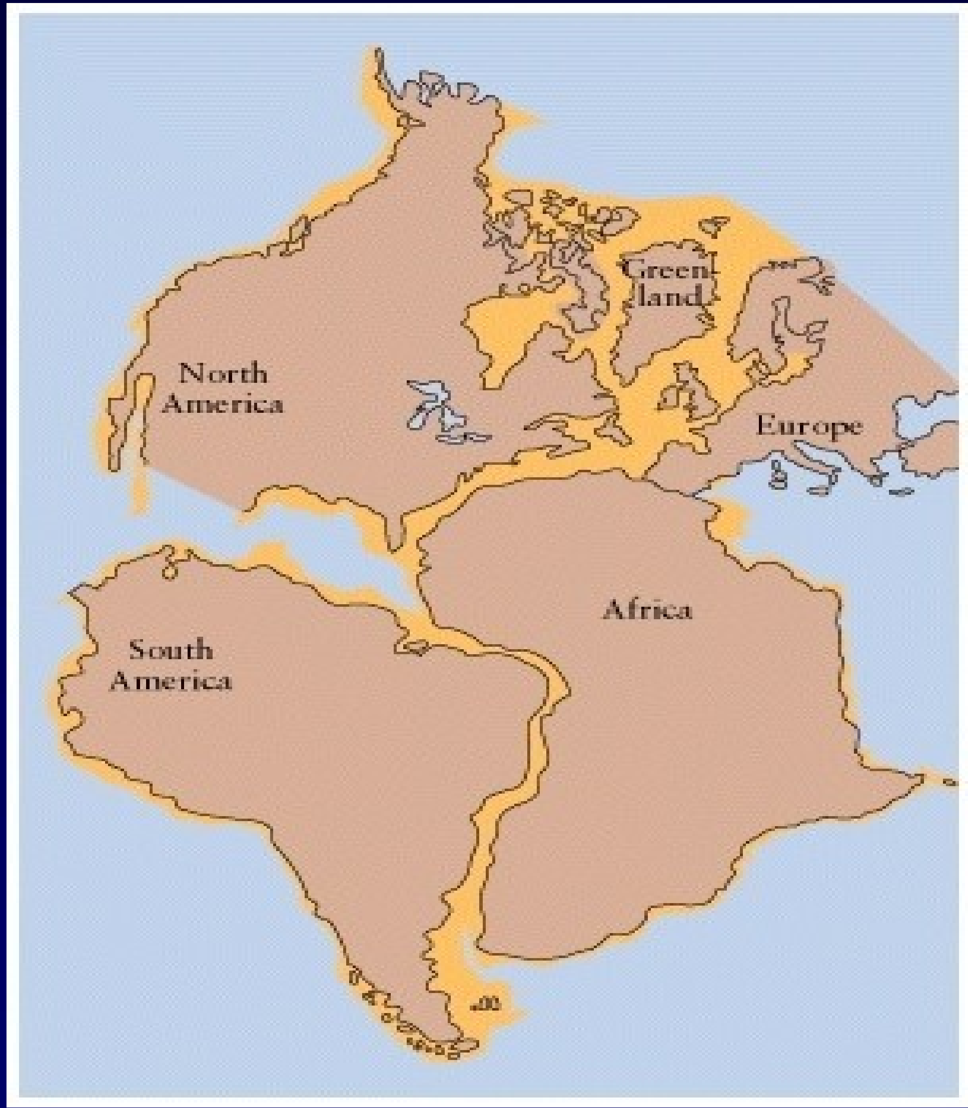
- All the continents were once combined into one super-continent called "Pangaea".
- Continents break up and gradually drift apart.
- Continents float on a denser underlying interior of the Earth.

- Wegner believed that continents were once compressed into a single protocontinent which he called Pangaea.
- He believed that Pangaea was intact until ~300 million years ago, when it began to break and drift apart.
- Over time they have drifted apart and presently occupy current position.

Evidences for Continental Drift

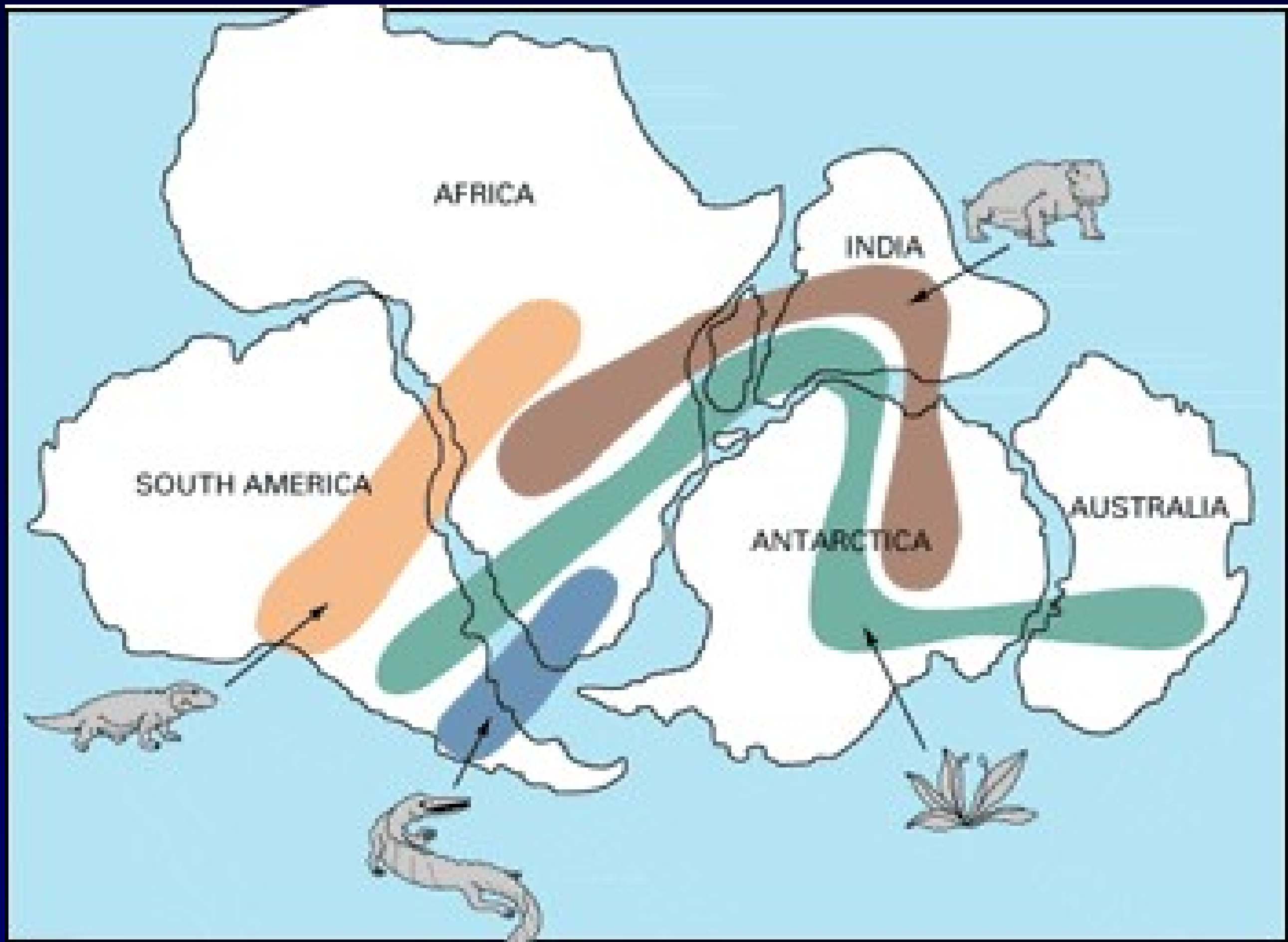
- Fit of continental coastlines
- Different continent have nearly identical fossils
- Continuity of structures and formations
- Widespread Paleozoic glaciation in southern continent

Continental Drift → Rocks Evidence



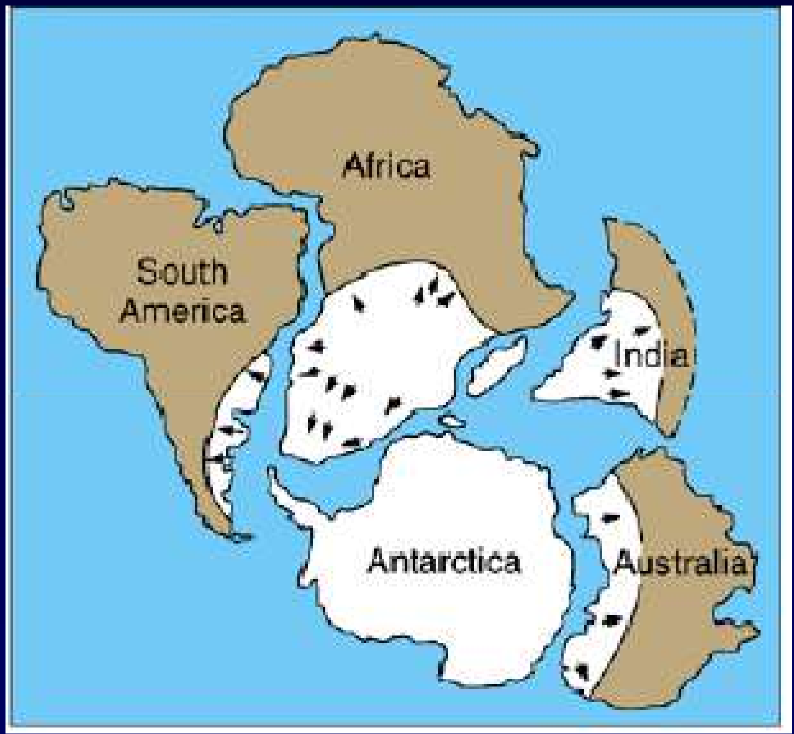
- Reasonably good matching rock types and fit of continental coastlines

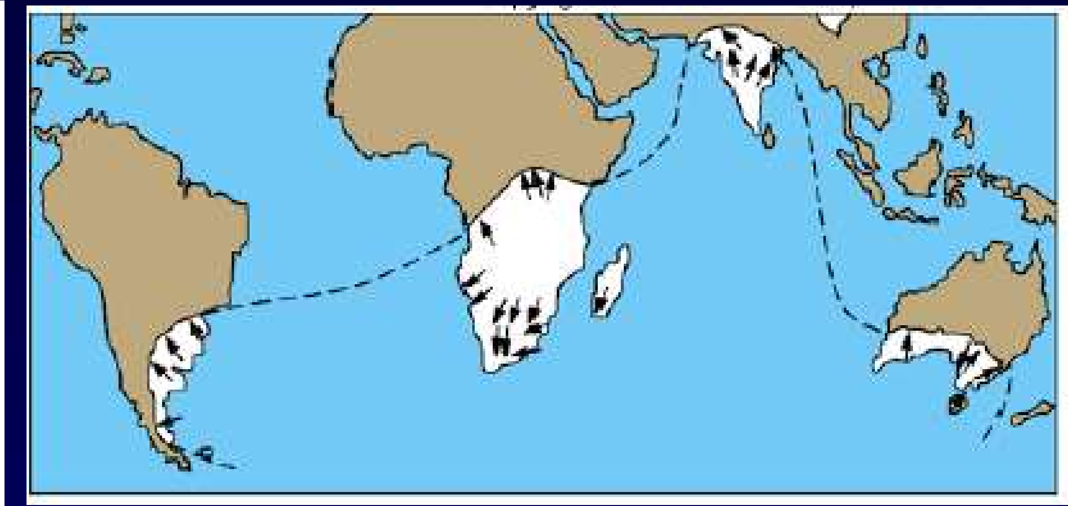
Continental Drift ➔ Fossils Evidence



- Plant and animal fossils, found on the coastlines of different continents, are nearly identical.

Continental Drift ➔ Evidence of glaciation





- Glaciation in Africa, South America, India, Australia and Antarctica during the same time

Dismissal of Theory of Continental Drift

- **Wegener had a great idea, and lots of supporting evidences.**
- **However, his theory was not widely accepted until the early 1970's.**

Why ?

Wegener's hypothesis lacked a geological mechanism to explain how the continents could drift across the earths surface.

Wegener's inability to provide an adequate explanation of the forces responsible for continental drift and the prevailing belief that the earth was solid, resulted in the scientific dismissal of his theories.

Drifting Continents & Shifting Theories

Arthur Holmes had made an attempt to explain drifting continents in 1931 based on concepts of continental drift and convection cells within the mantle.

- He believed that the mantle undergoes thermal convection.
- This idea is based on the fact that as a substance is heated its density decreases and rises to the surface until it is cooled and sinks again.
- This repeated heating and cooling results in a current which may be enough to cause continents to move.
- This idea received very little attention at the time. Why ? → Due to lack of evidences.

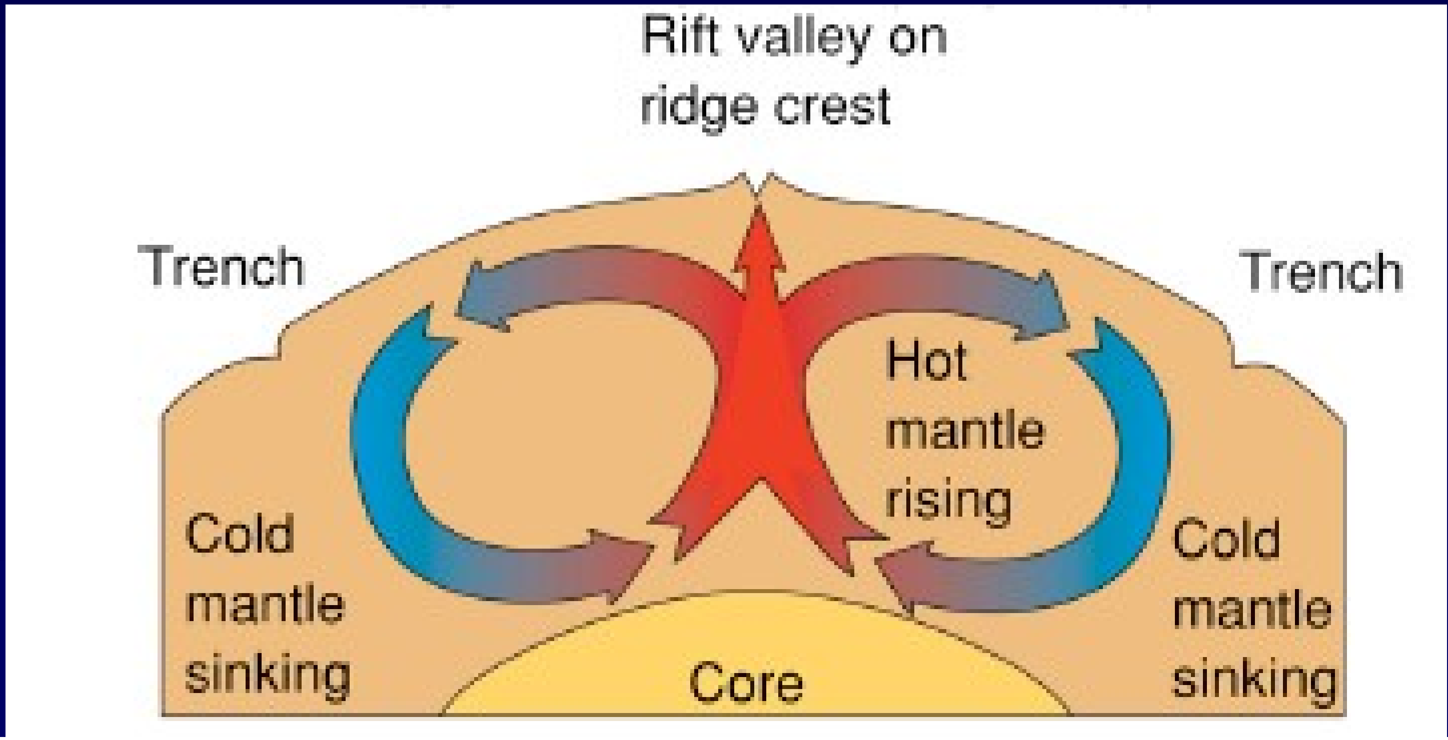
Seafloor Spreading & Plate Tectonics

(Second step in development of plate tectonics)

- 1961-1962 ➔ Harry Hess & R. Deitz
- Mantle Convection Currents Model, now known as Seafloor Spreading
- This idea was basically the same as proposed by Arthur Holmes ~30 years earlier.
- The theory had much more evidences to further develop and support the idea.
- Greater understanding of the ocean floor, geomagnetic anomalies parallel to the mid-oceanic ridges, and discovery of mid-ocean ridges, oceanic trenches etc.

Seafloor Spreading

- Convection cells: Mantle upwells under MOR
- New oceanic crust is formed at MOR, then spreads laterally. High elevation, heat flow, basaltic volcanism
- Oceanic crust is dragged down at trenches
- Continents ride passively between sites of upwelling and down welling



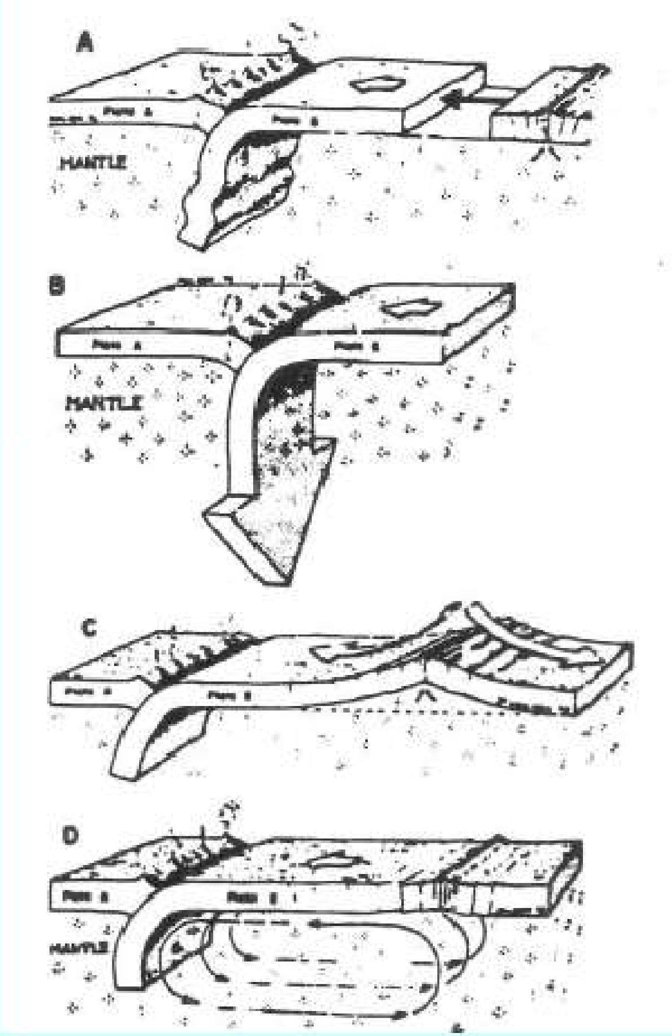
Critical Concept of Seafloor Spreading

- reversals of Earth's magnetic field
- acquisition of magnetism in rocks
- presence of mid-ocean ridges
- bathymetry of sea-floor
- composition of oceanic crust
- convection in mantle (heat loss from interior)

Driving Forces of Plate Tectonics

Immediate mechanism is more controversial: 4 possibilities . . .

- **Pushing** from ridges➔ compressional stress in plate
- **Pulling** by downgoing slab➔ tensional stress in plate
- Gravity **sliding** from height of MOR to abyssal plain/trench
- **Dragging** by convection cells acting on base of the plate

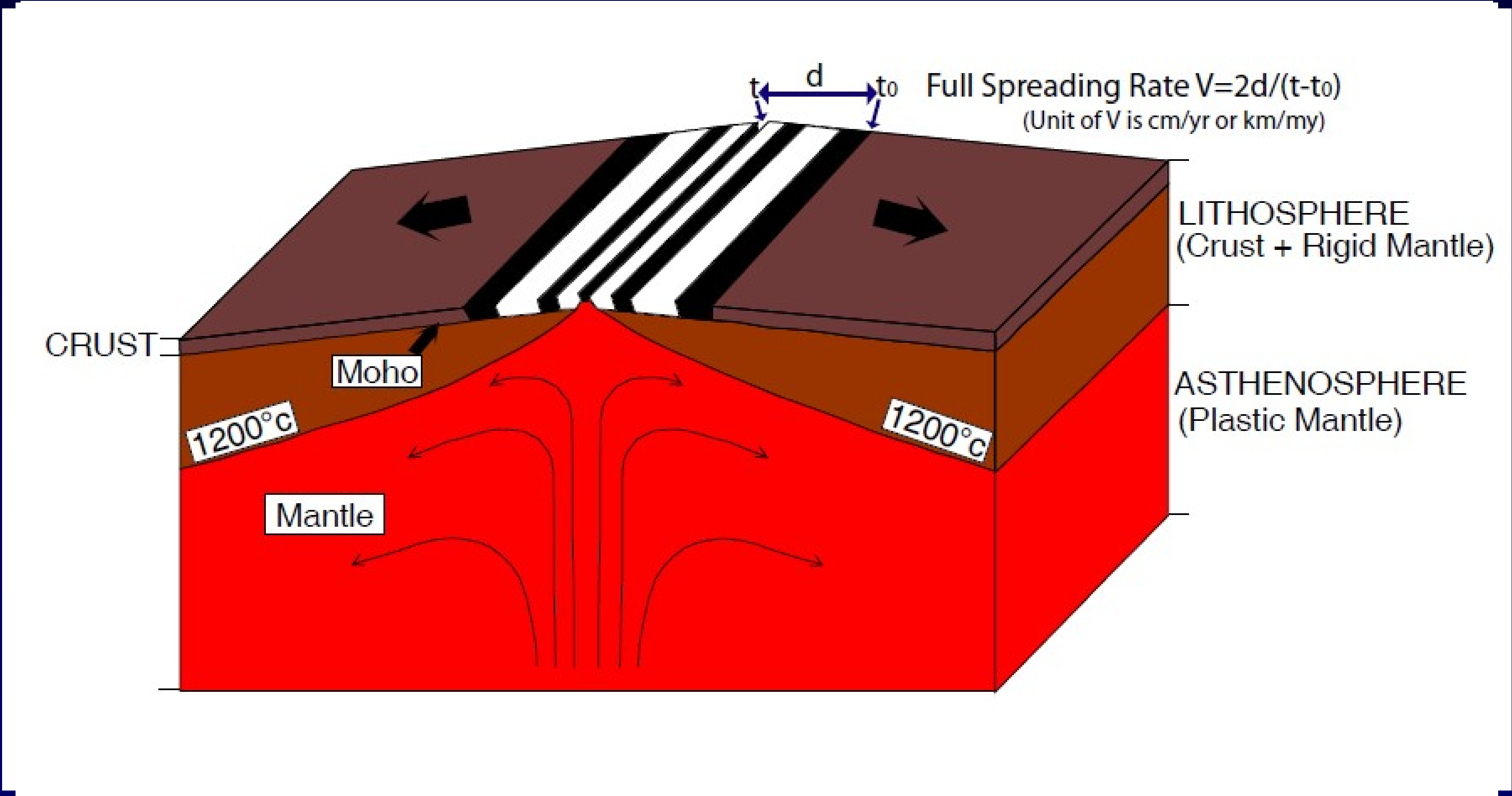


What causes continents to drift?

Convection Currents

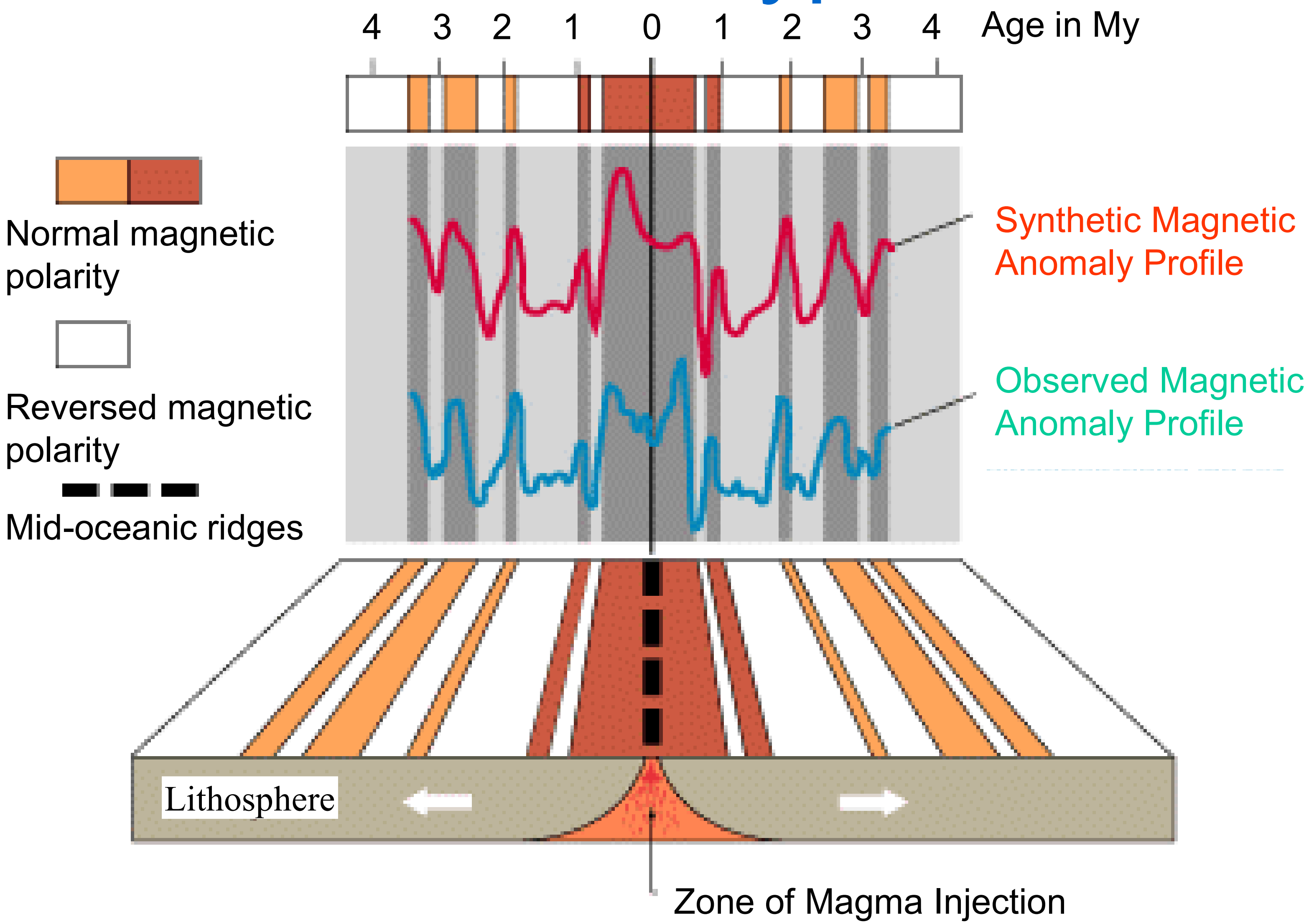
- Hot magma in the Earth moves toward the surface, cools, then sinks again.
- Creates convection currents beneath the plates that cause the plates to move.

Seafloor Spreading



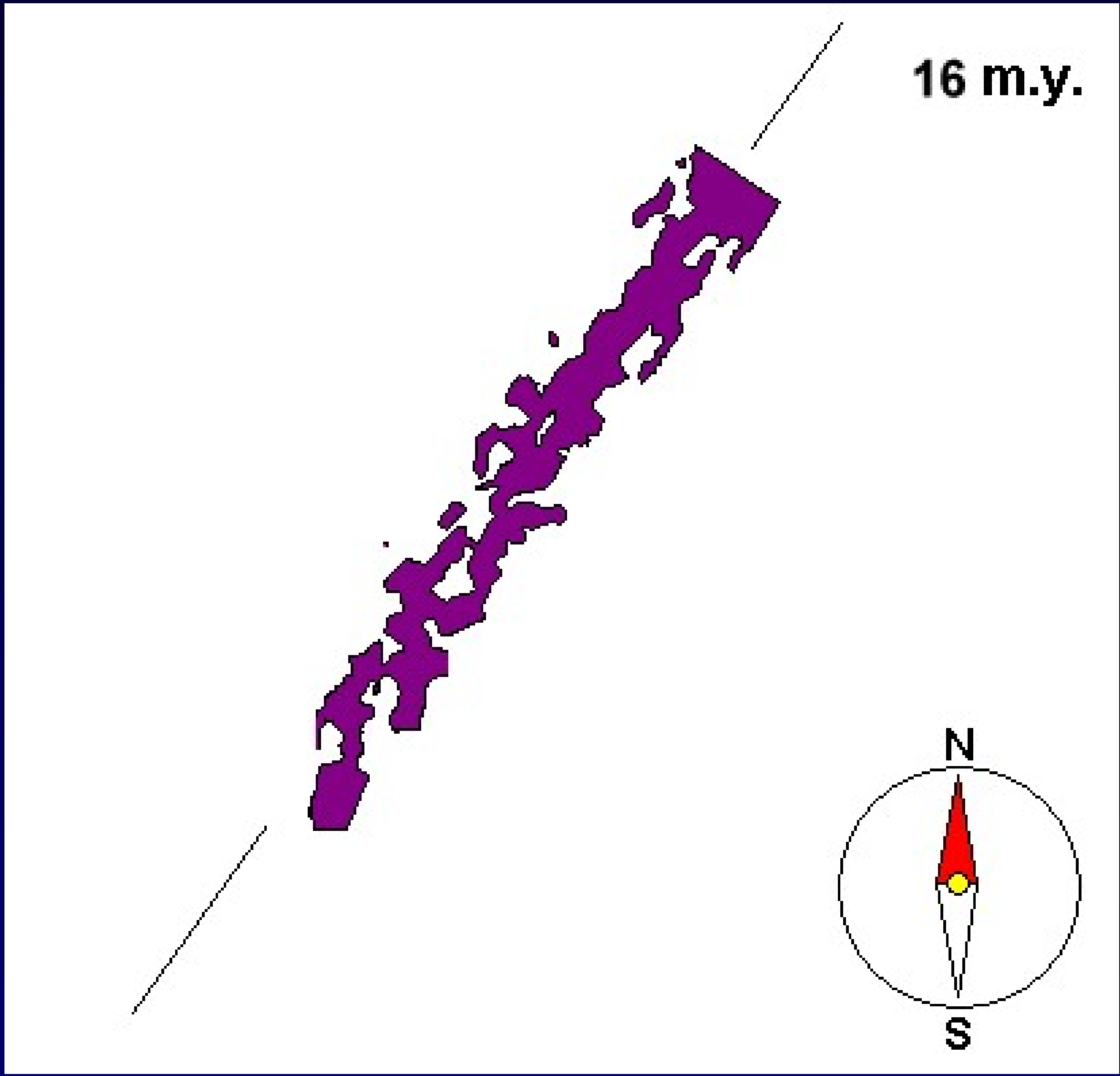
- Plates move away from each other due to mantle convection
- Mantle upwells hot material under MOR -- high elevation, heat flow, basaltic volcanism

Vine Matthews Hypothesis (1963)



The Hypothesis ➔ Seafloor Spreading and Reversals of the Earth's Magnetic Field

Sea-Floor Spreading



Further supports of Seafloor Spreading Hypothesis

- International Geophysical Year (1957-58)
- International Indian Ocean Expedition (1959-65)
- Heezen and Tharp (1965)
- DSDP expedition (1965)
- ODP Expedition (1985) ➔ JOIDES (Joint Oceanographic Institutes for Deep Earth Sampling) resolution (1985)
- Canadian geologist John Tuzo Wilson (1908-1993)
- D.P. Mackenzie and R. L. Parker

and several others ……………..

Basic Concepts of Plate Tectonics

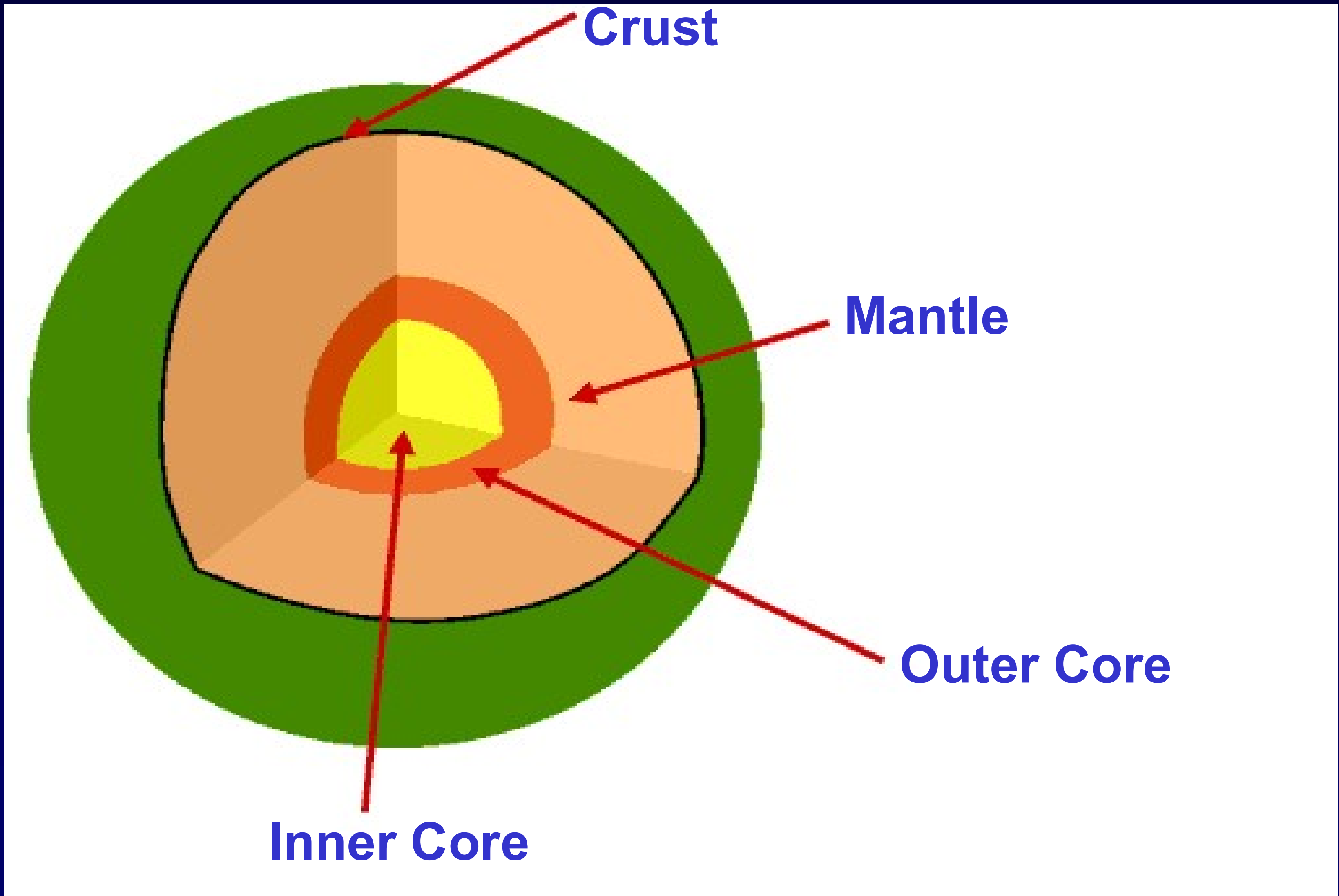
Interior Structure of the Planet Earth

- A globally-average value of radius of the Earth is usually considered as 6,371 km, although it is about 21 km larger at equator than at poles. Its equatorial radius is ~6378 km, but its polar radius is ~6357 km.

- Density (mass/volume), Temperature, and Pressure increase with depth in the Earth.

- The Earth has a layered structure. This layering can be viewed in two different ways (i) layers of different chemical composition and (ii) layers of differing physical properties.

1. Internal Structure of the Earth based on Chemical Composition



1. Internal Structure of the Earth based on Chemical Composition

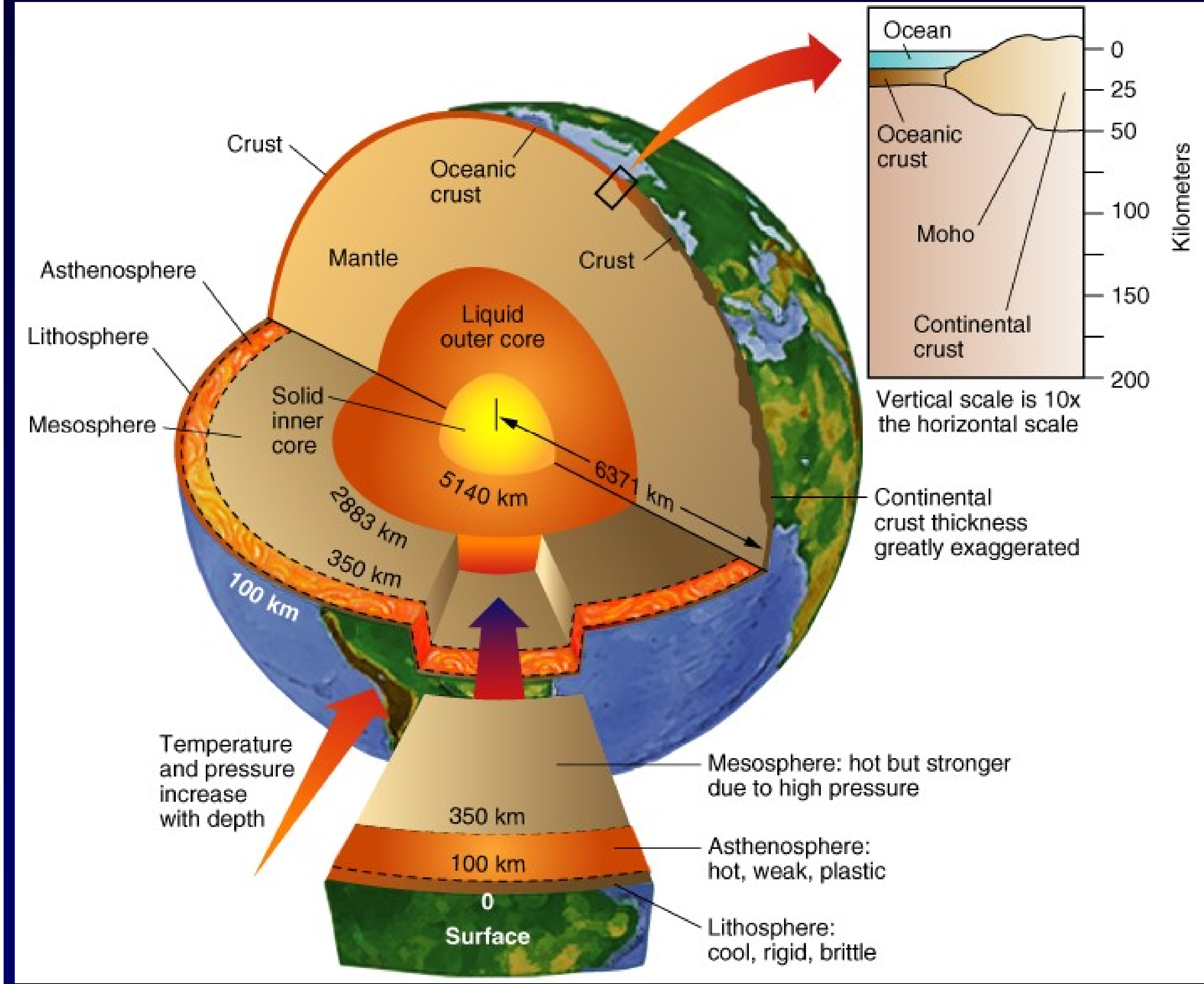
TYPE	THICKNESS	TEMPERATURE	MATERIAL
CRUST	5 – 35 km	N/A	Rock minerals
MANTLE	2900 km	200-4000º C	Iron, magnesium and aluminium
OUTER CORE	2250 km	2730-4230º C	Molten iron + Nickel
INNER CORE	1230 km	5430º C	Solid iron + Nickel

ER= ~6378 km, PR= ~6357 km

2. Internal Structure of the Earth based on Physical Properties

- **Lithosphere** - about 100 km thick, very brittle, easily fractures at low temperature.
- **Asthenosphere** – about 250 km thick - solid rock, but soft and flows easily (ductile).
- **Mesosphere** - about 2500 km thick, solid rock, but still capable of flowing.
- **Outer Core** - 2250 km thick, Fe and Ni, liquid
- **Inner core** - 1230 km radius, Fe and Ni, solid

Gross Structure of the Earth



Oceanic Crust

Basaltic
SiMa
Denser
Thinner
Mostly below sea Surface
Much younger in age

Continental Crust

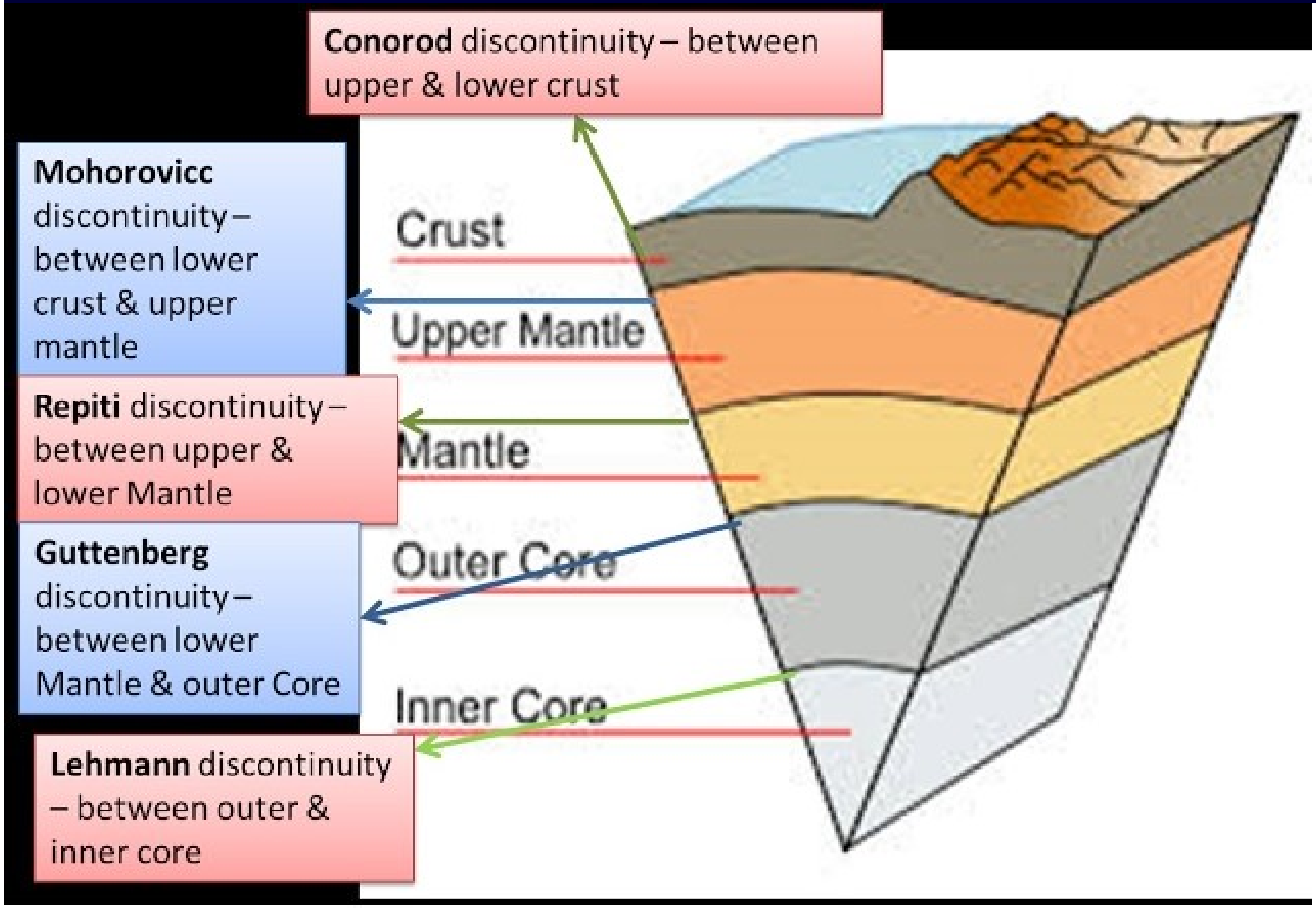
Granitic
SiAl
Lighter
Thicker
Mostly above sea Surface
Much older in age

Discontinuities / Transition Zones inside the earth

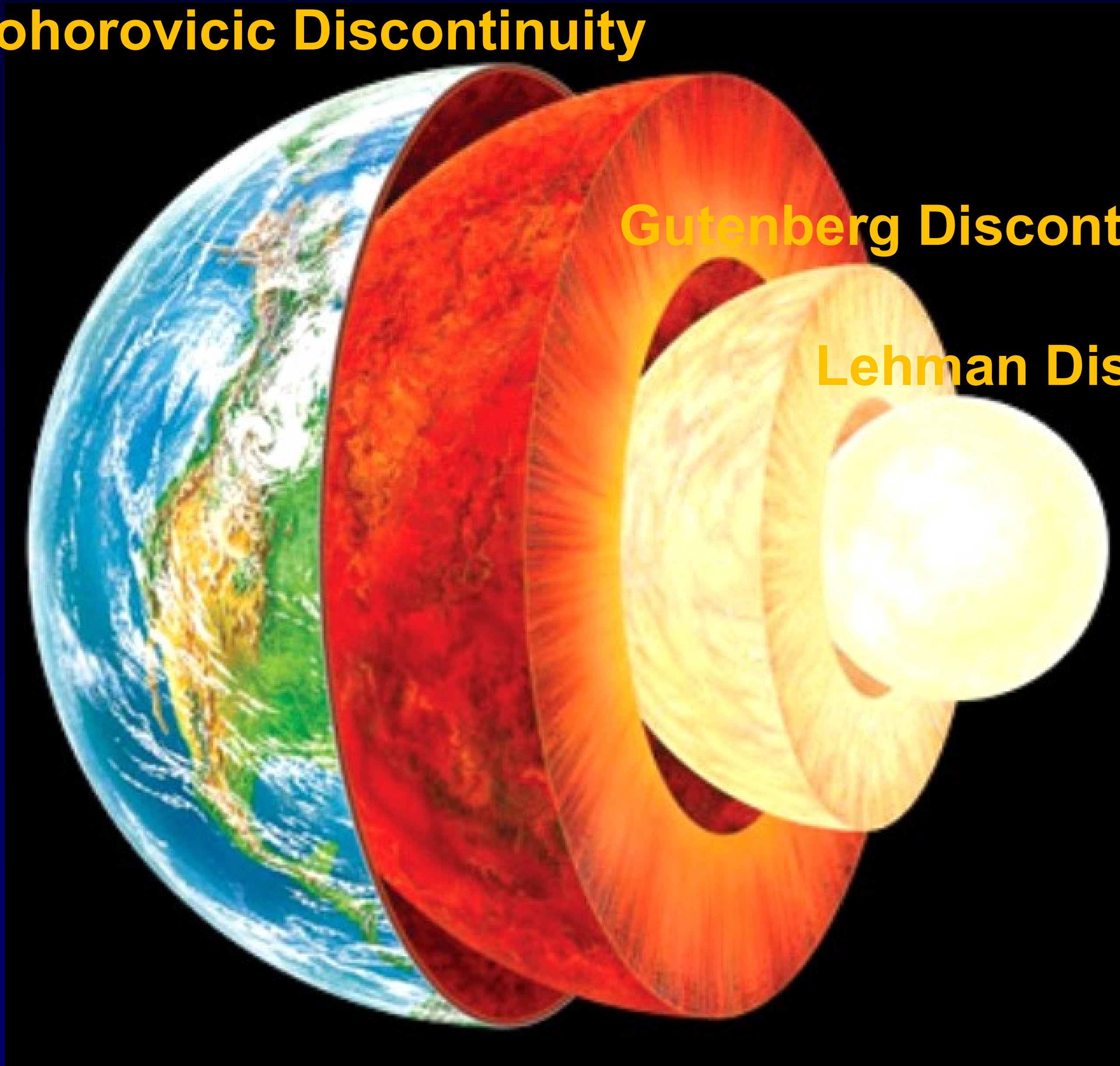
Unique layers within the Earth are separated from each other through a transition zone. These transition zones are called discontinuities. They are marked by clear-cut variations in density of material, velocity of seismic waves, temperature and pressure conditions. There are five discontinuities/ transition zones inside the earth:

1. **Conrad Discontinuity** (named after the seismologist Victor Conrad)**:** Transition zone between upper (SiAl) and lower Crust (SiMa).
2. **Mohorovicic Discontinuity** (named after Andrija Mohorovicic): Transition zone between the Crust and Mantle.
3. **Repitti Discontinuity** (named after William C. Repetti): Transition zone between Outer mantle and Inner mantle.
4. **Gutenberg Discontinuity** (named after Beno Gutenberg): Transition zone between Mantle and Core.
5. **Lehman Discontinuity** (named after seismologist Inge Lehmann): Transition zone between Outer core and Inner core.

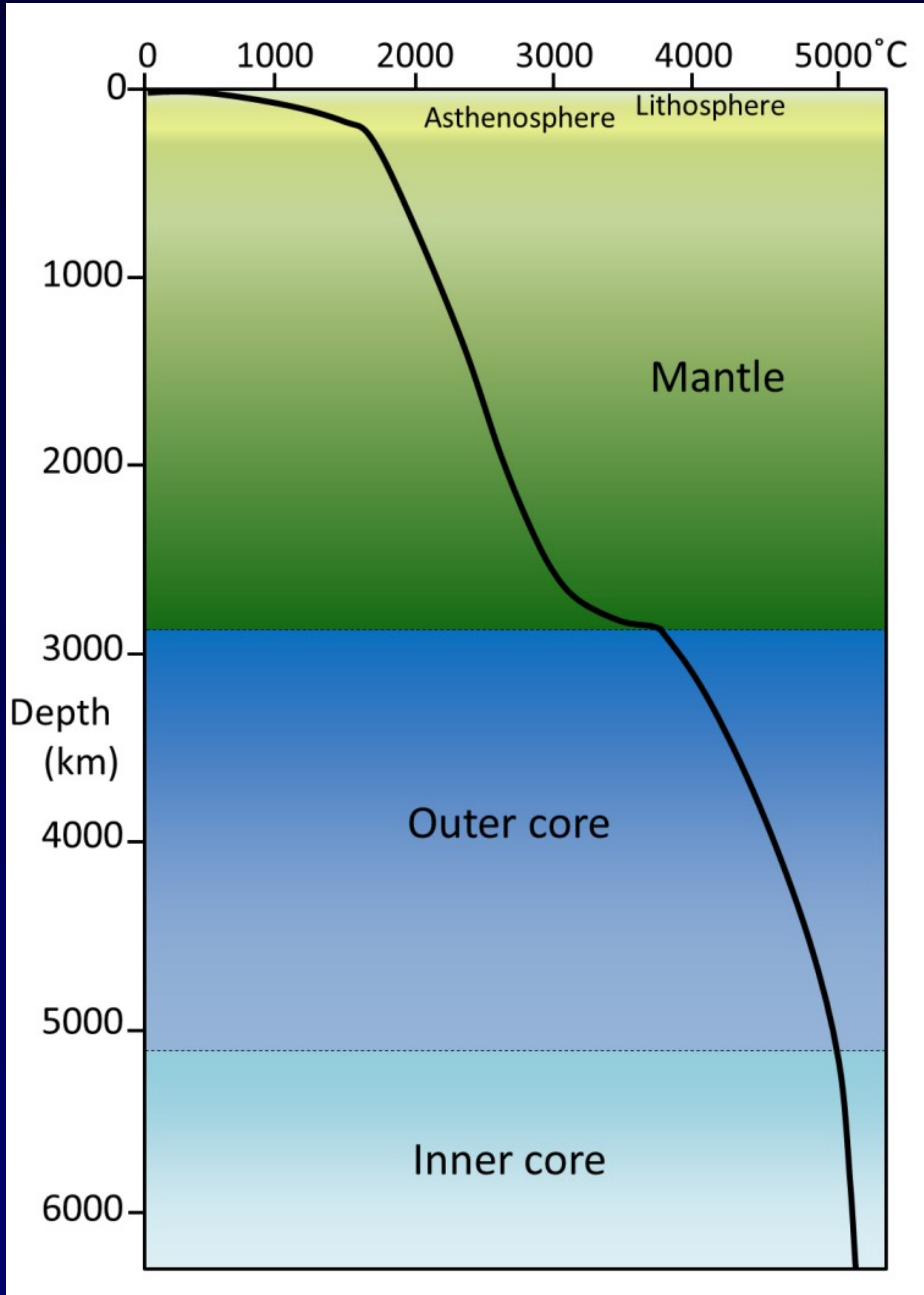
Discontinuities / Transition Zones inside the earth



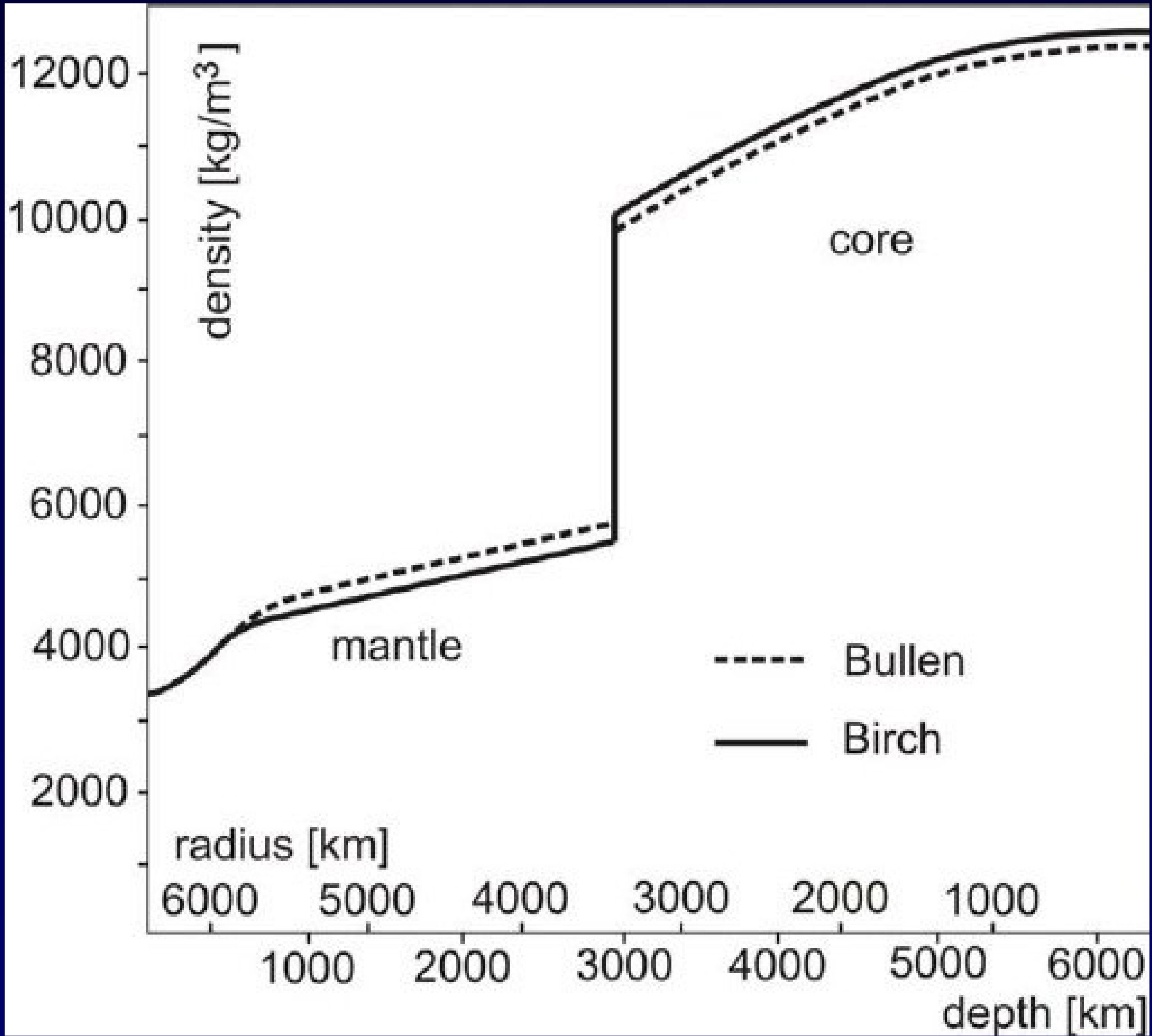
Major Discontinuities inside the earth



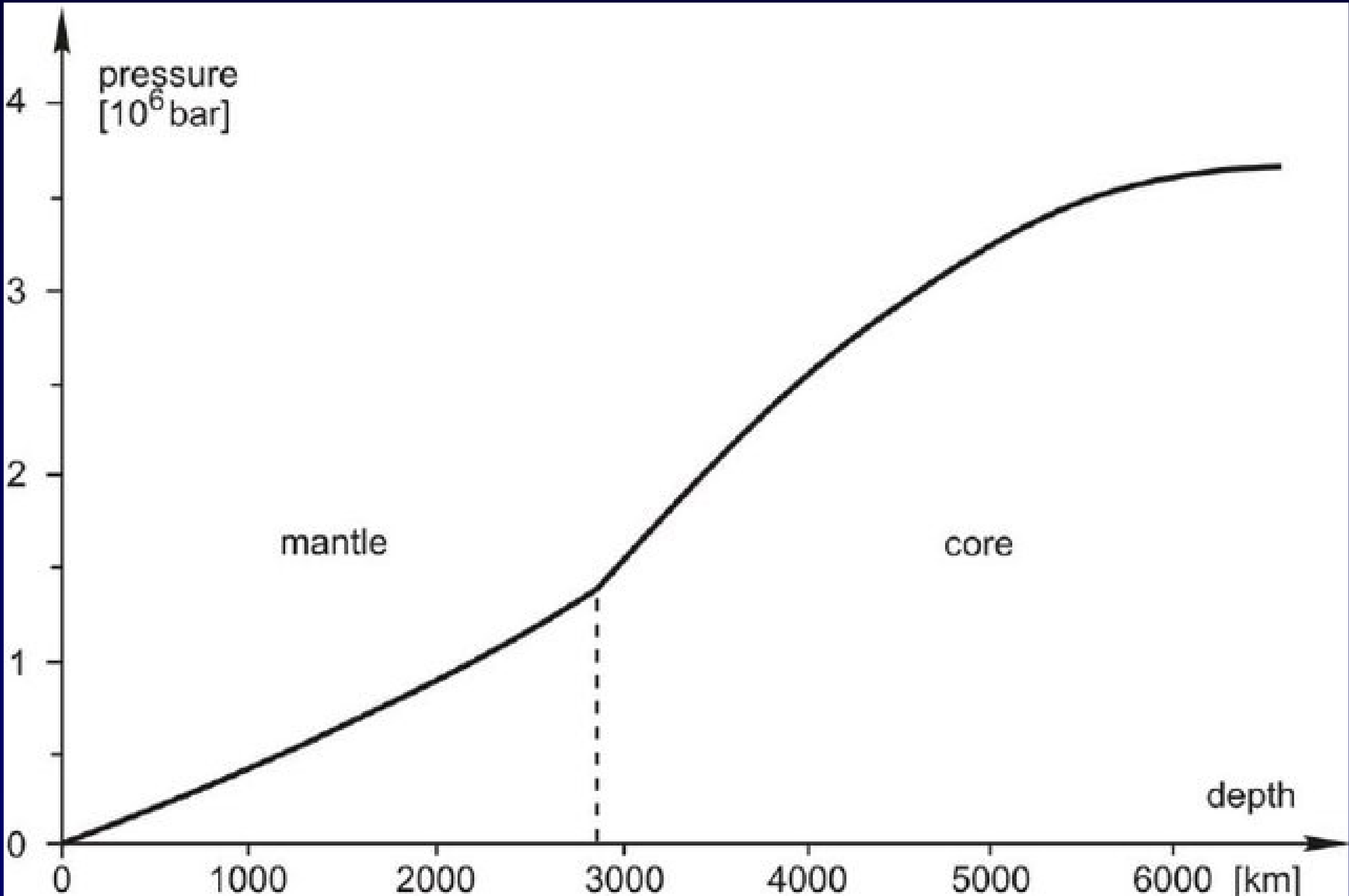
Temperature Variation with Depth



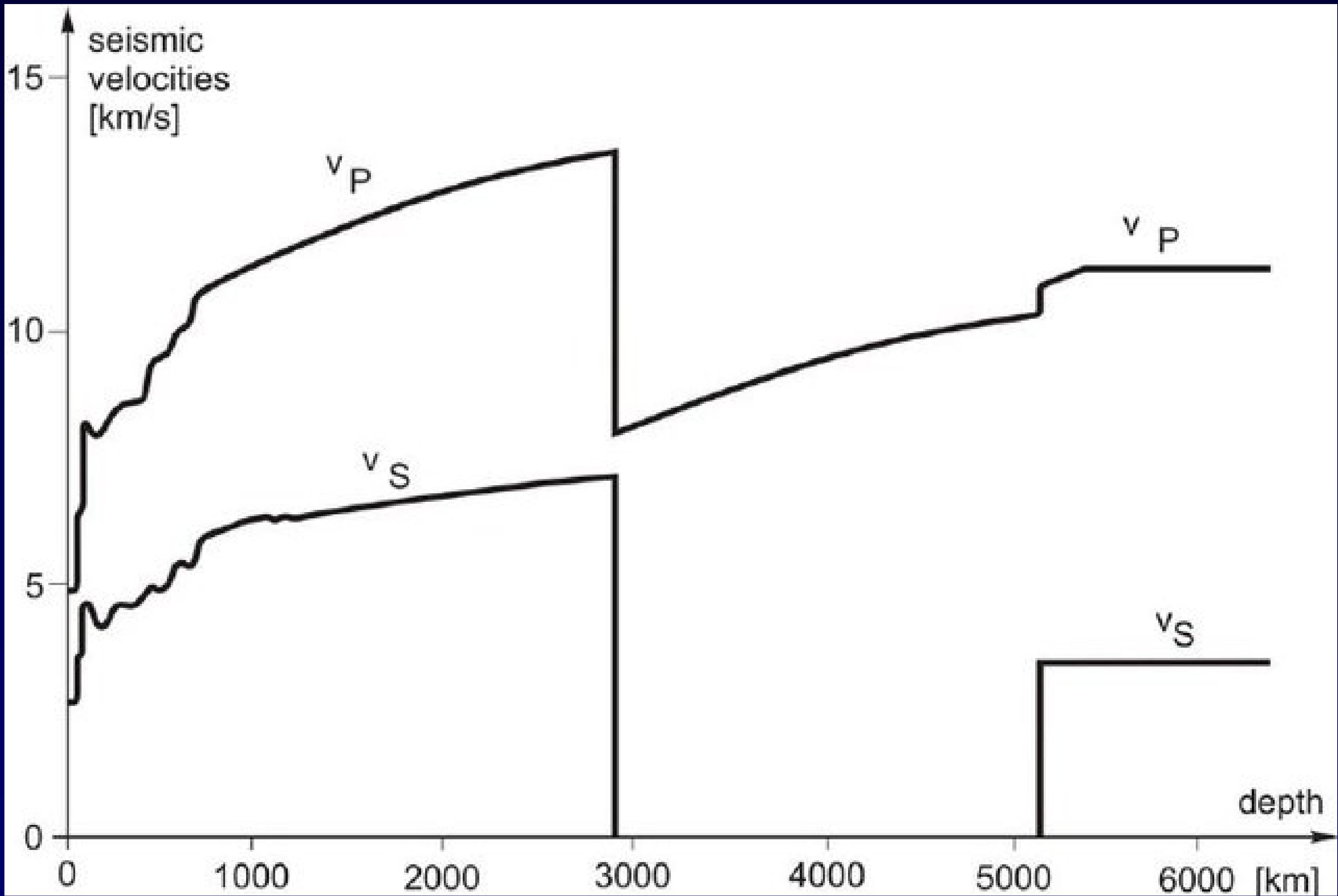
Density Variation with Depth



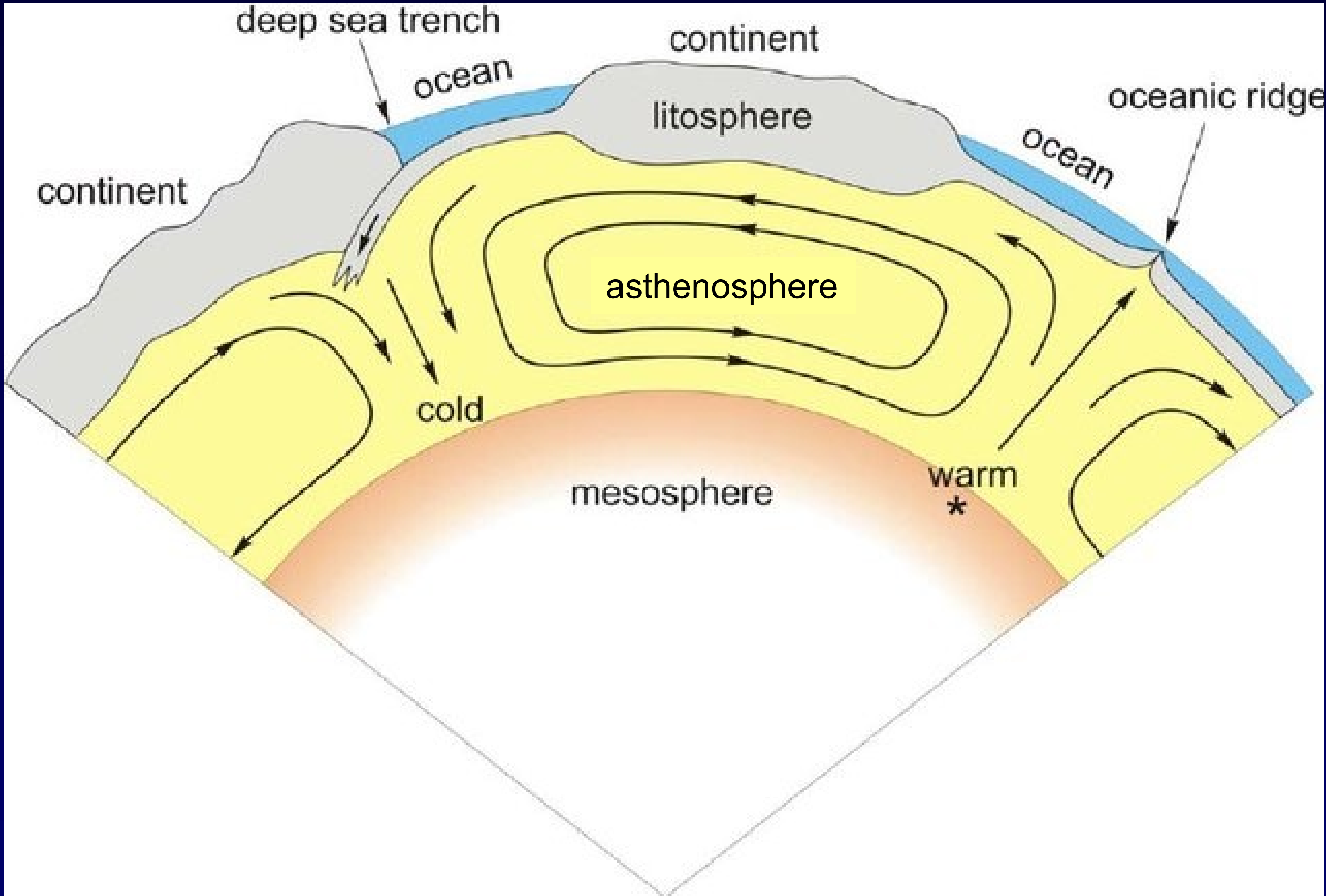
Pressure Variation with Depth



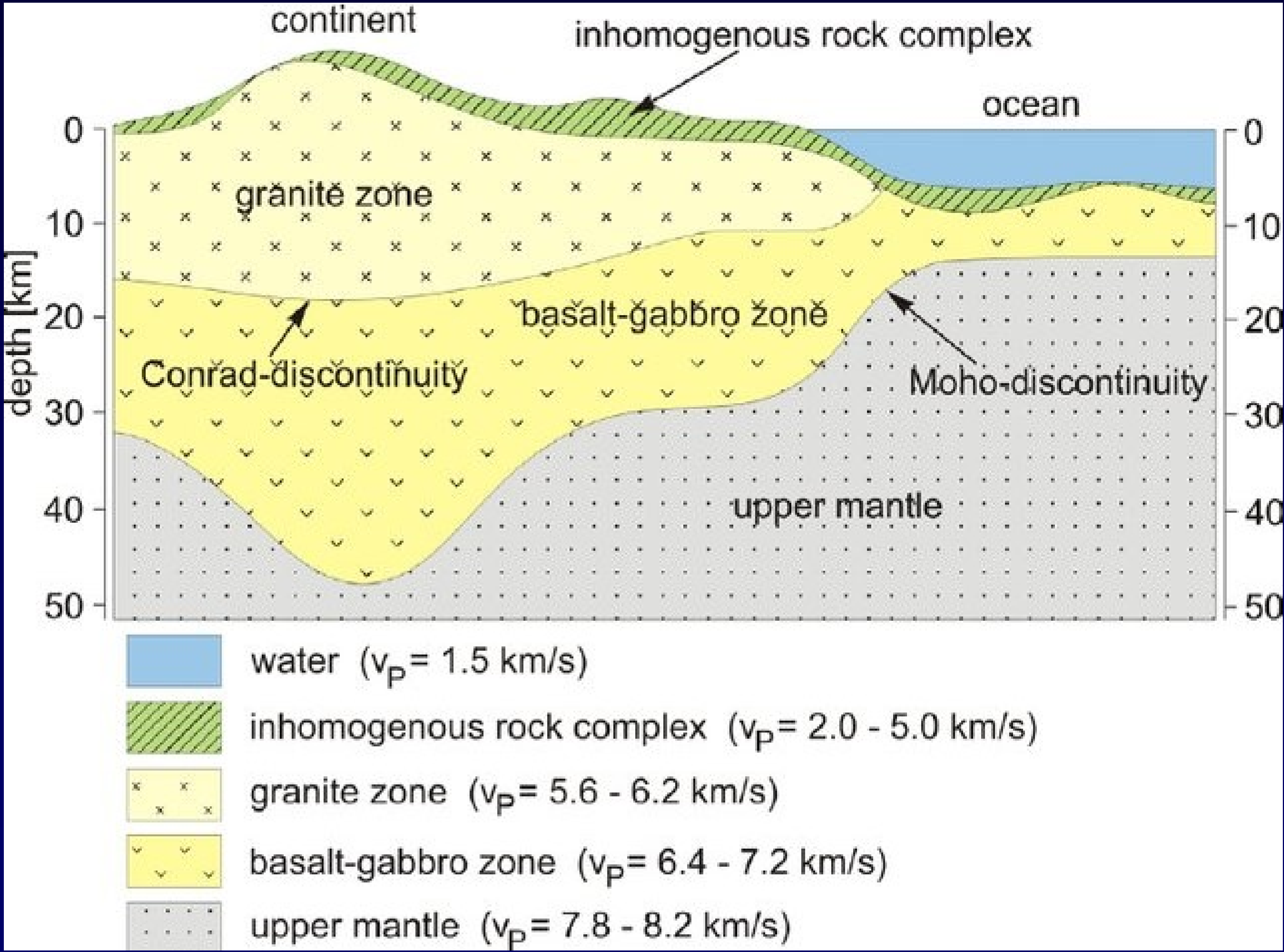
Seismic Velocity Variation with Depth



Thermal Convection in the Mantle

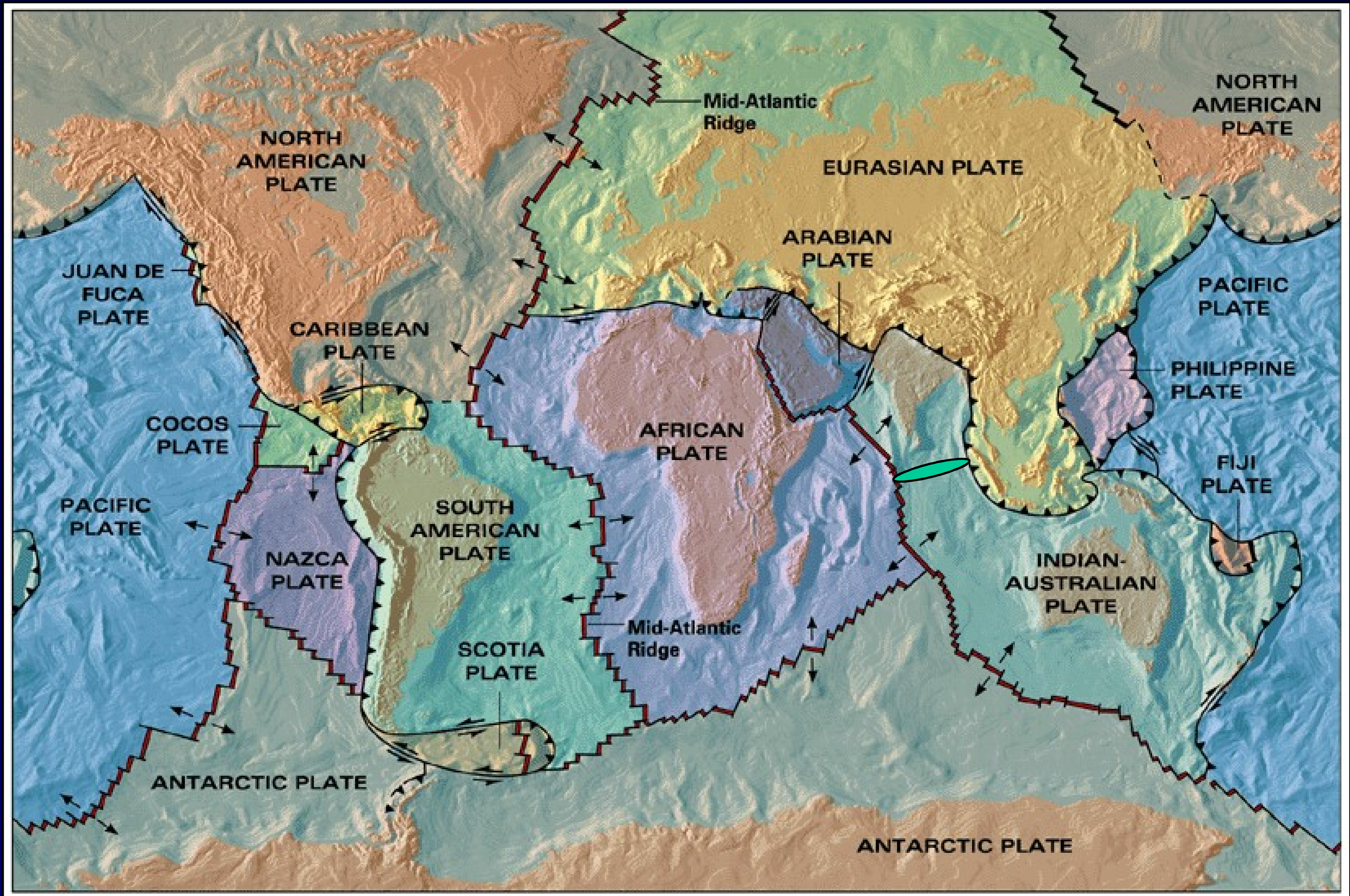


Structure of Continent and Ocean crusts



Tectonic Plates

[Tectonic Plates ➔ Outer rigid layer (~70-100 km thick) of the earth]

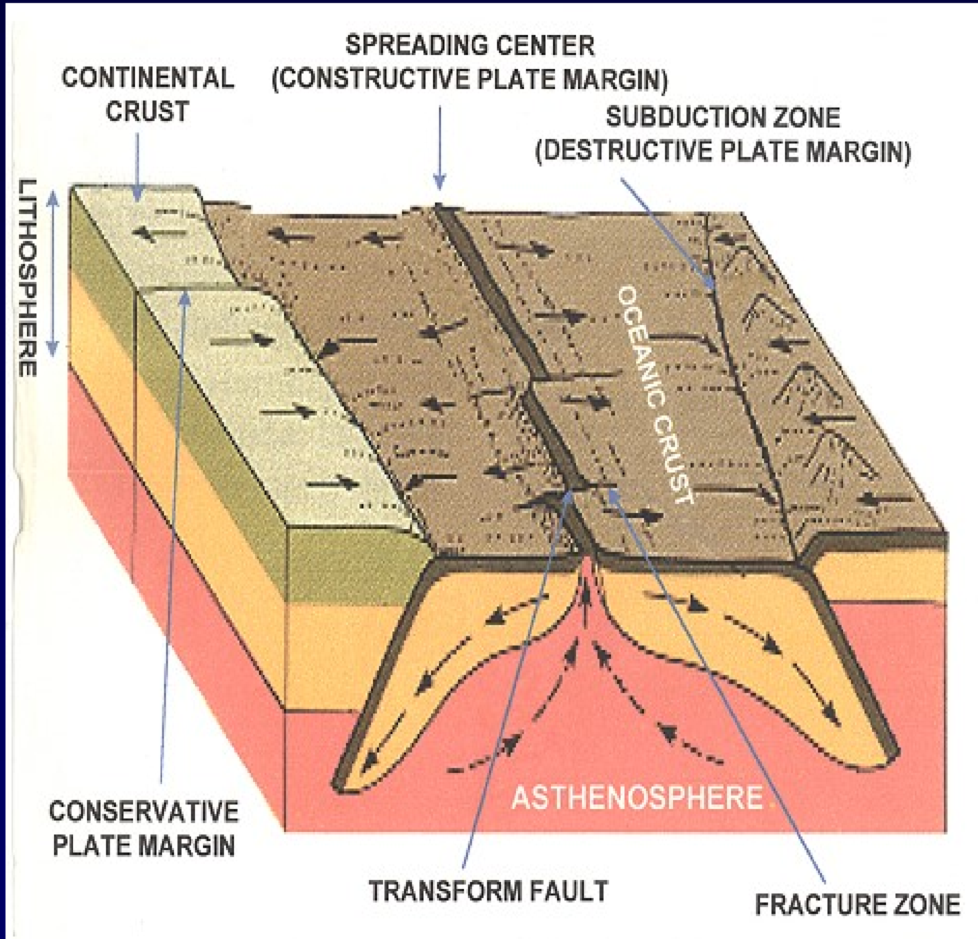


Divergent Plate boundary   Convergent Plate boundary   Transform Plate boundary

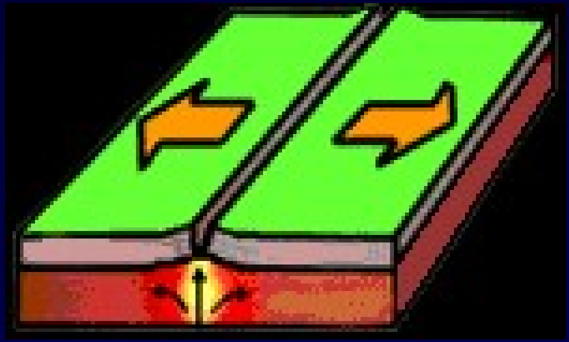
- The plates move slowly at a speed of few centimeters per year over the Asthenosphere.
- Asthenosphere is highly viscous, easily deformable layer between upper and lower Mantle.
- Most volcanic and earthquake activities occurs at or near plate boundaries.
- The place where the two plates meet is called a plate boundary.
- Plate interiors are relatively quiet.

Plate Margins / Boundaries

- Divergent
- Convergent
- Transform
- Plate Boundary Zone

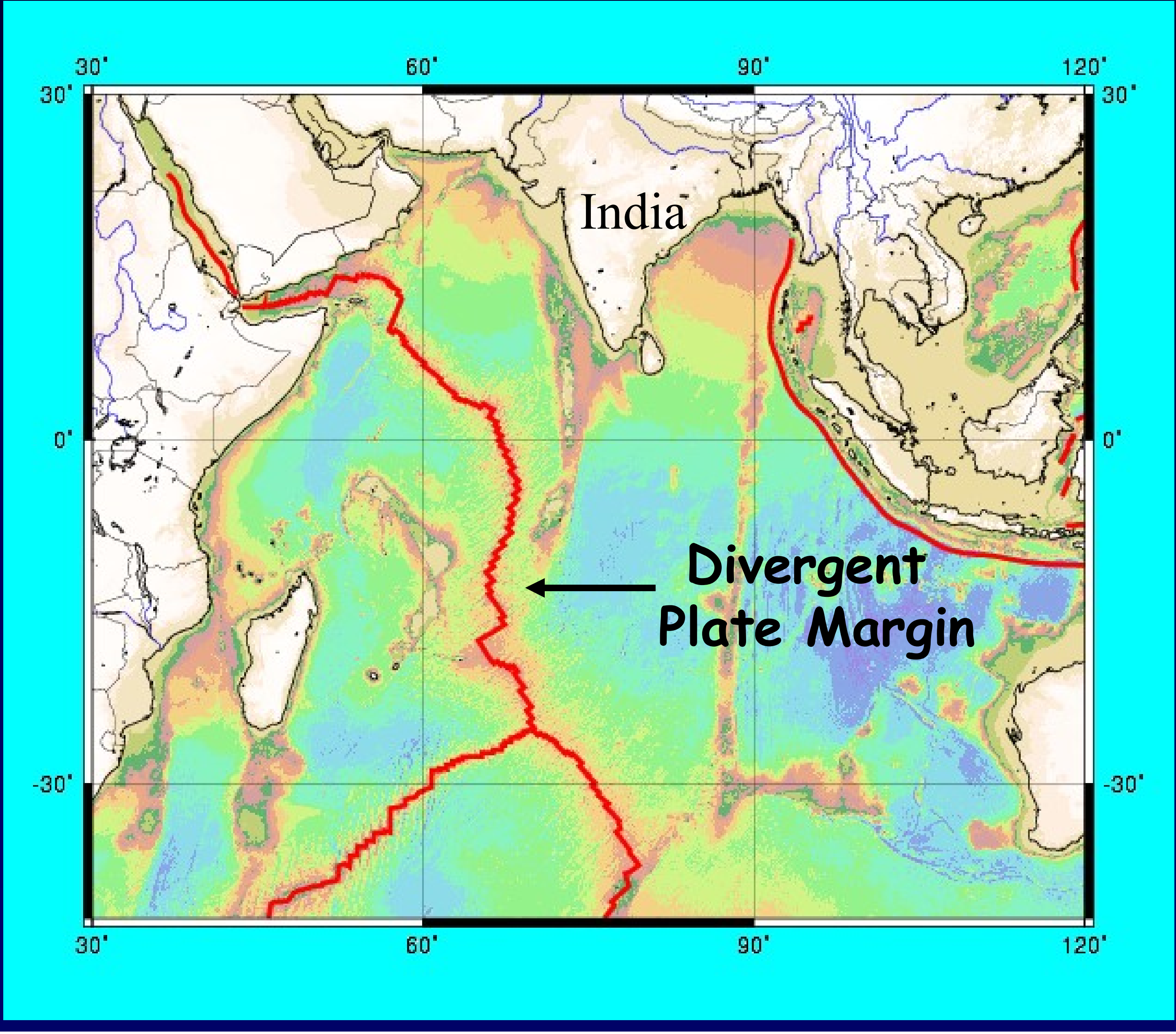


Divergent Plate Margins



- Plates move apart
- This pulling apart causes Seafloor Spreading
- Volcanic activity is always present
- Earthquakes are shallow
- Two types of boundaries:
  - (i) Oceanic
  - (ii) Continental

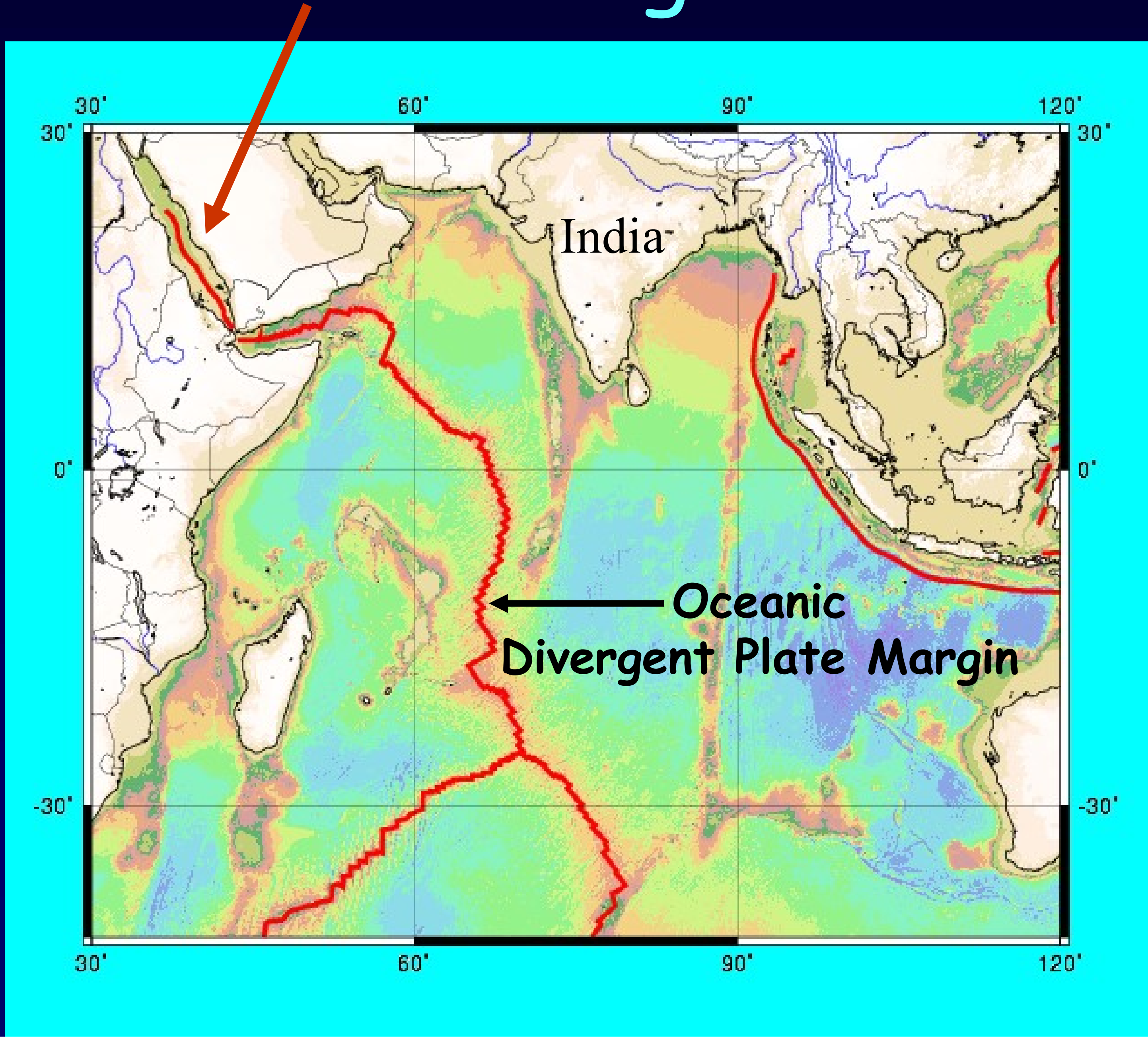
1. Oceanic Divergent Margin



2. Continental Divergent Margin

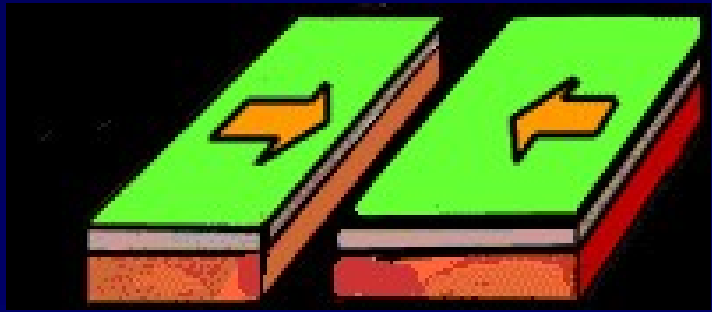
- Continental rifting
- Eventually, oceanic crust is formed and divergence becomes oceanic type

Continental Divergent Boundary



Convergent Plate Margins

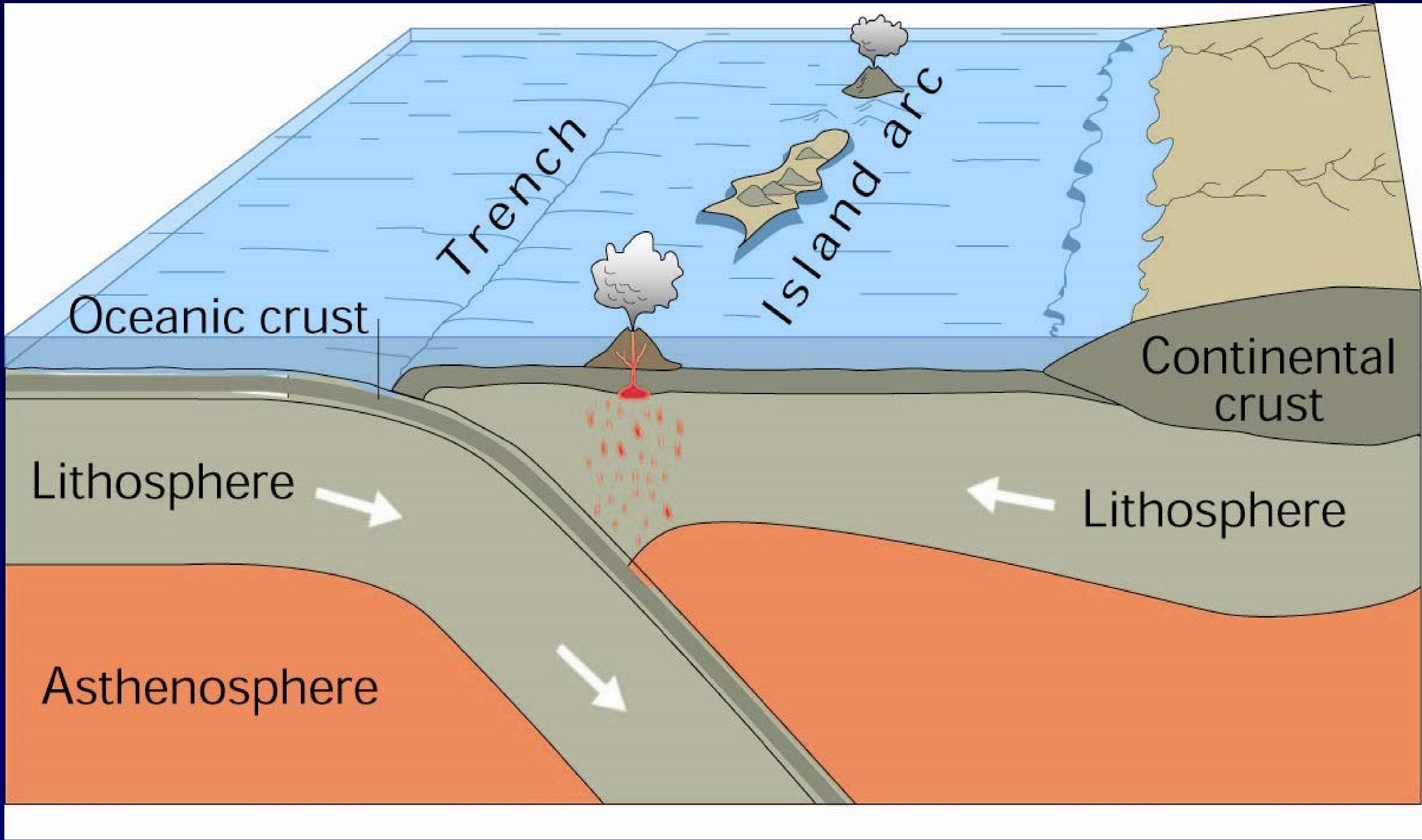
- Plates move toward each other
- Volcanic activity common
- Deep, medium, and shallow earthquakes
- Three types of Convergent Margin
    1. Ocean-ocean
    2. Ocean-continent
    3. Continent-continent



1. Ocean-ocean Convergence

- Subduction – one oceanic plate subducts (dives) beneath the other
- Volcanic island arc is formed
- Trenches are found which is the deepest parts of the ocean floor and are created by subduction

1. Ocean-ocean Convergence

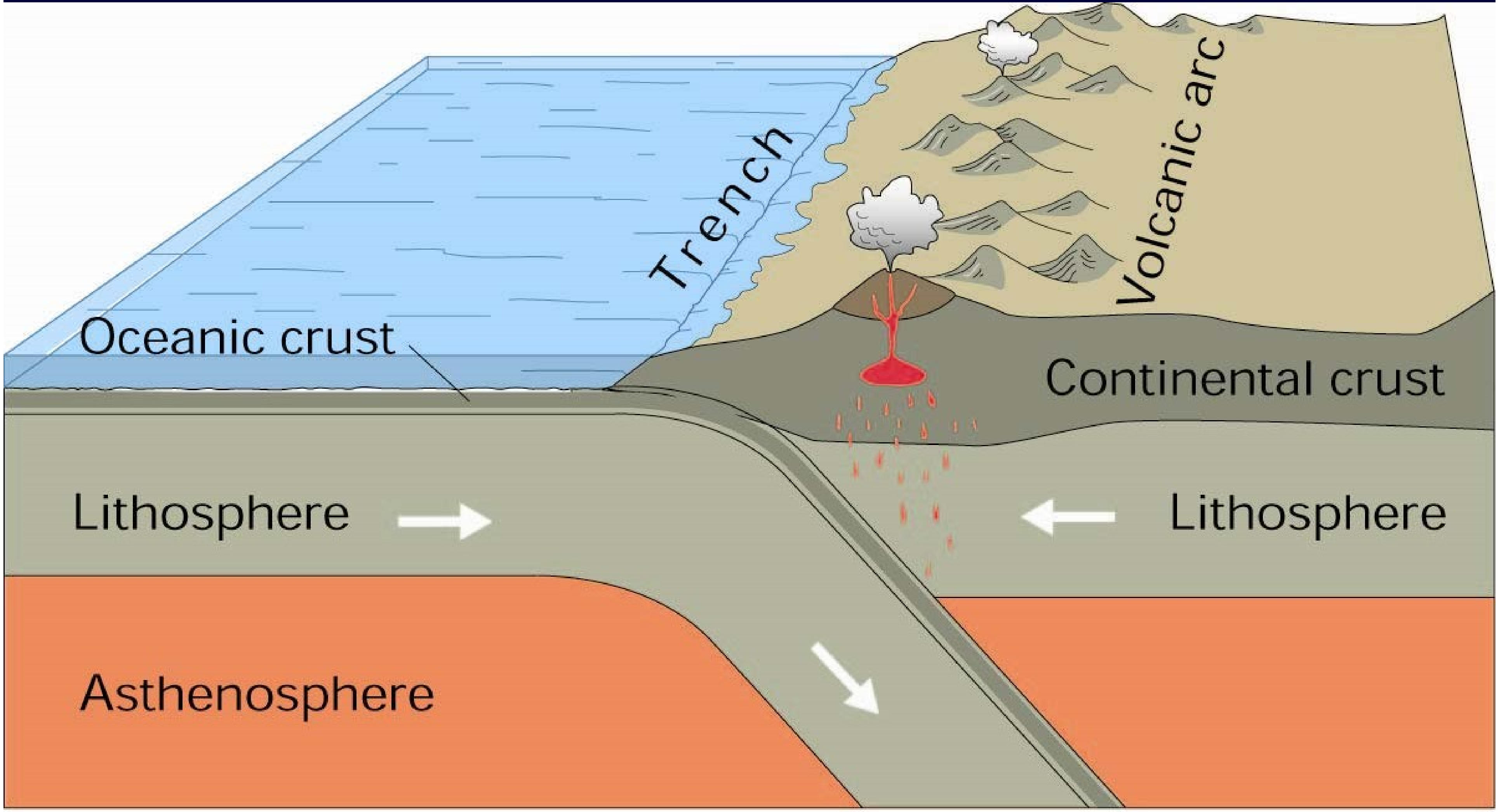


2. Ocean-continent Convergence

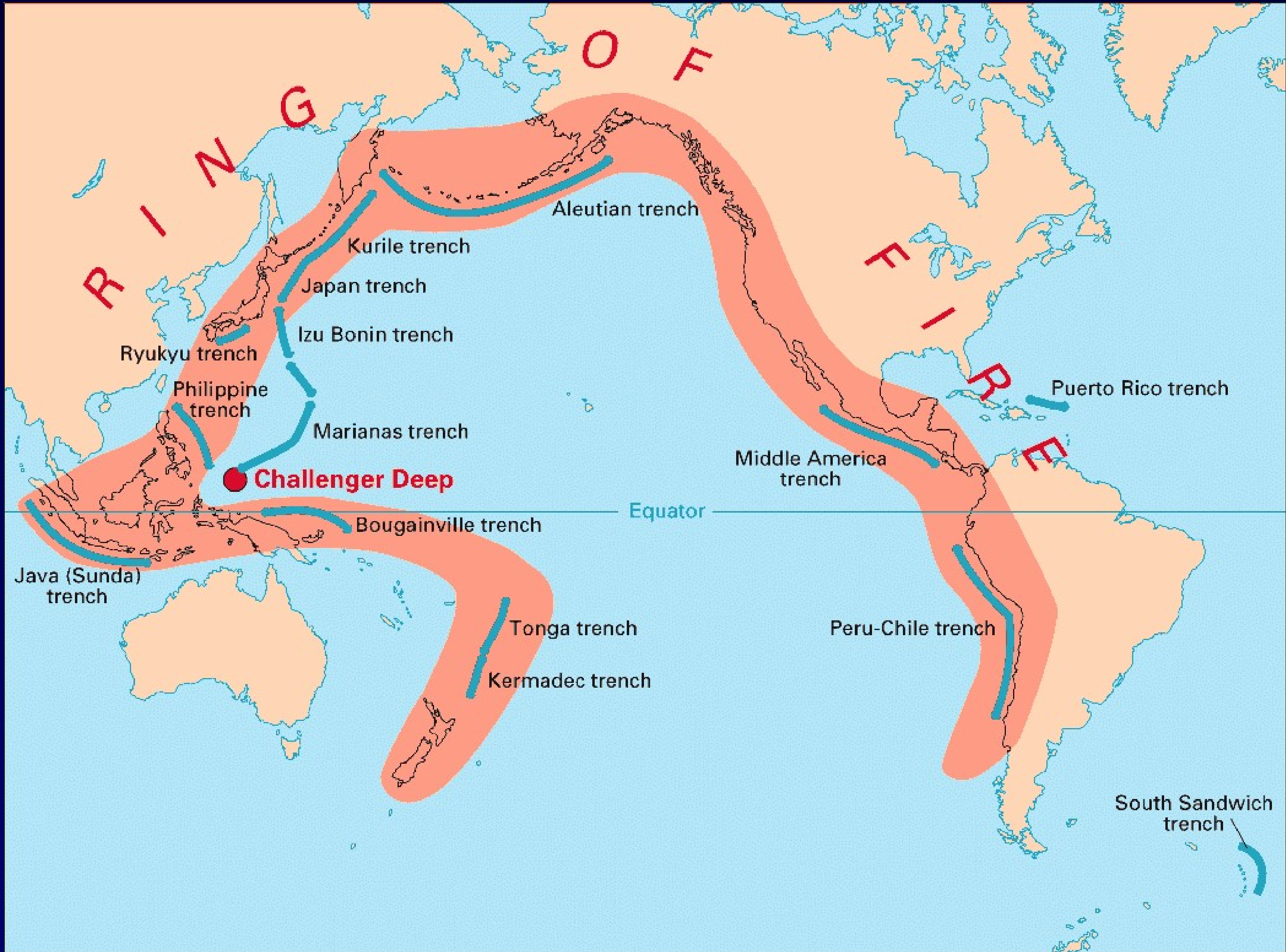
- During ocean-continent convergence, oceanic lithosphere always subducts beneath continental lithosphere

- Continental volcanic arc formed

2. Ocean-continent Convergence

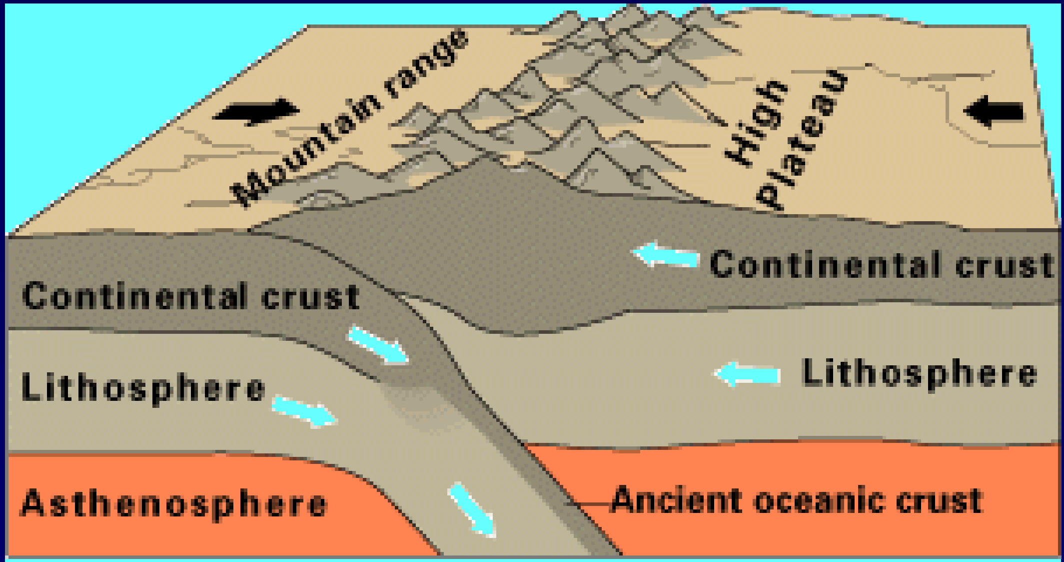


2. Ocean-continent Convergence



3. Continent-continent Convergence

- Subduction does not occur (continental crust is too buoyant to be subducted)
- Mountain building occurs without volcanism

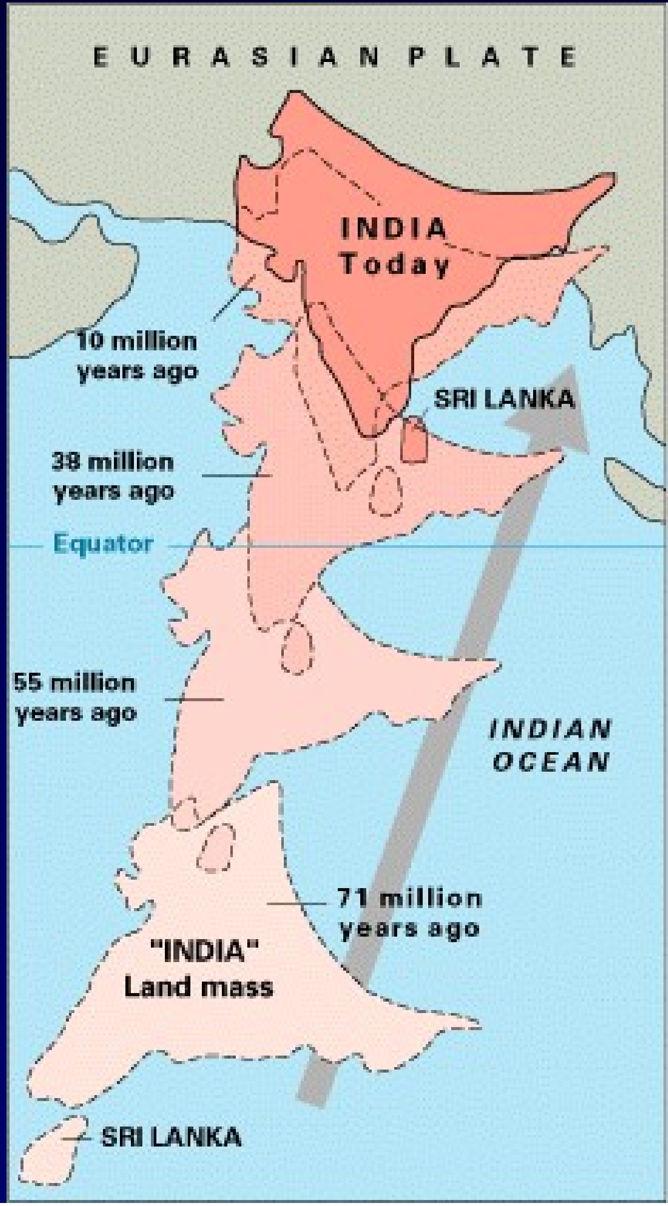


Continental-continental convergence

3. Continent-continent Convergence

India-Asia Collision

- Collision of Indian Plate with Eurasian plate caused uplift of the Himalaya ➔ the highest mountains on the Earth.

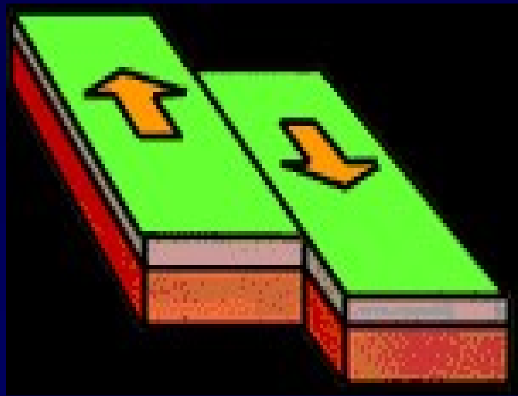


Subduction Zone and Oceanic Trenches

- A subduction zone is an area of tectonic plate collision where the more dense plate subducts beneath the less dense plate.
- Subduction zones create geologic formations such as mountain ranges, ocean trenches, and island arcs, as well as phenomena like earthquakes and volcanoes.
- A trench is a long depression in the ocean crust which is deeper than it is width, and the width is significantly smaller than its length.
- Ocean trenches are formed due to subduction of dense tectonic plate beneath the less dense tectonic plate and some of them may be extremely deep.

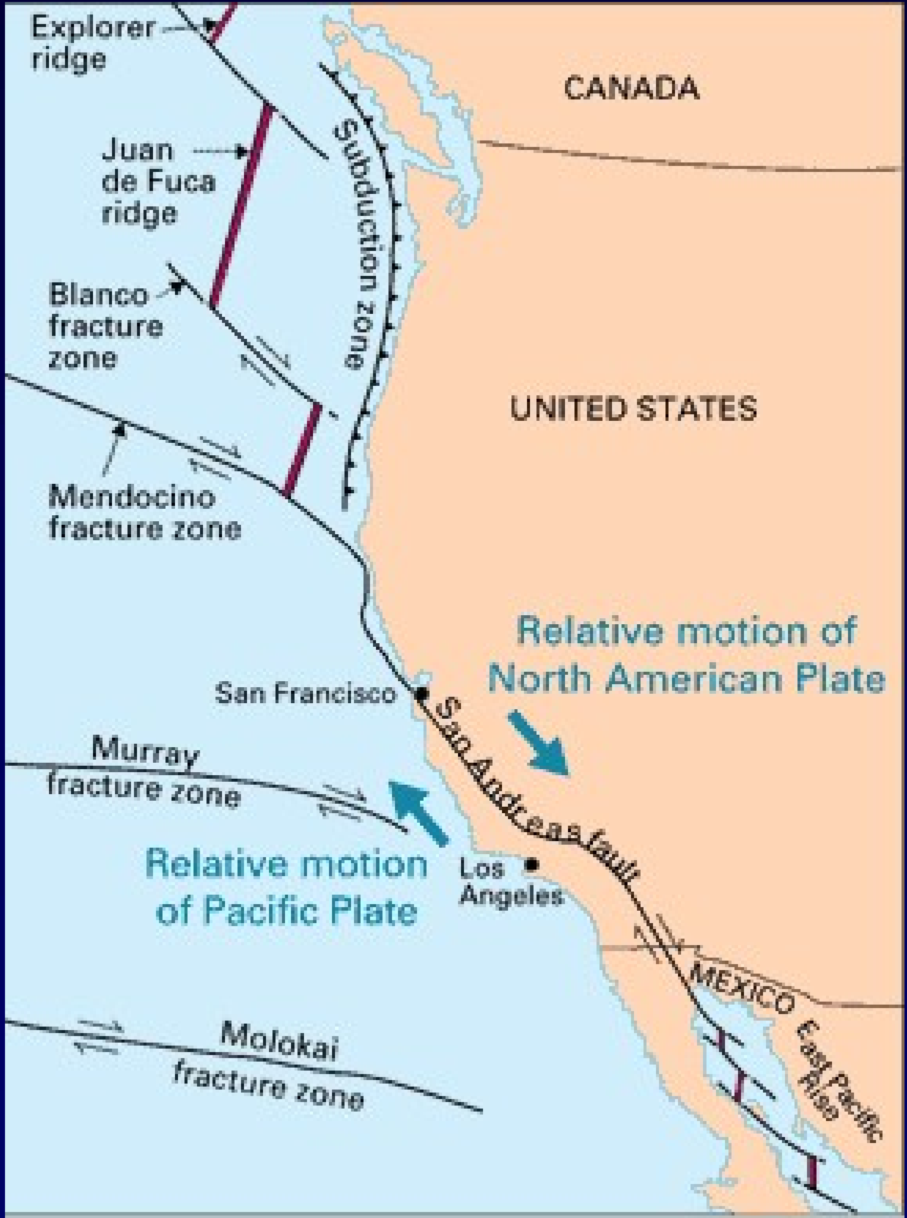
Transform Plate Margins

- Plates slide past each other ➔ New crust neither created nor destroyed
- Little or no volcanic activity
- Shallow earthquakes



- Tranform faults ➔ active
- Fracture zones ➔ inactive extensions of transform fault - "fossil transforms"

Transform Plate Margin



Pacific plate is moving NW relative to the North American Plate

As plates move, transform fault slips, and earthquakes occur

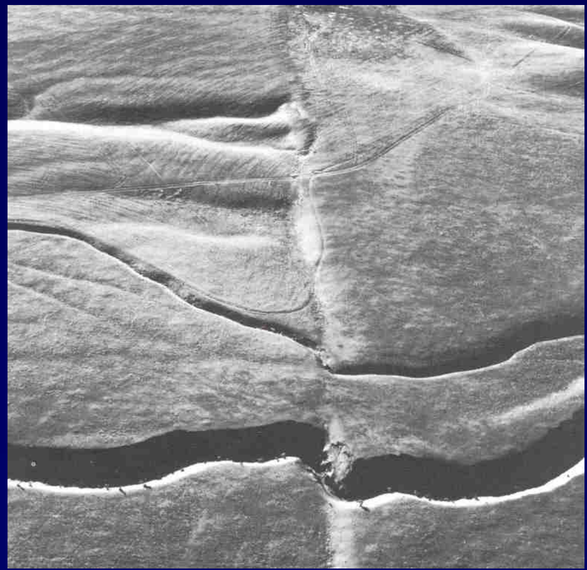
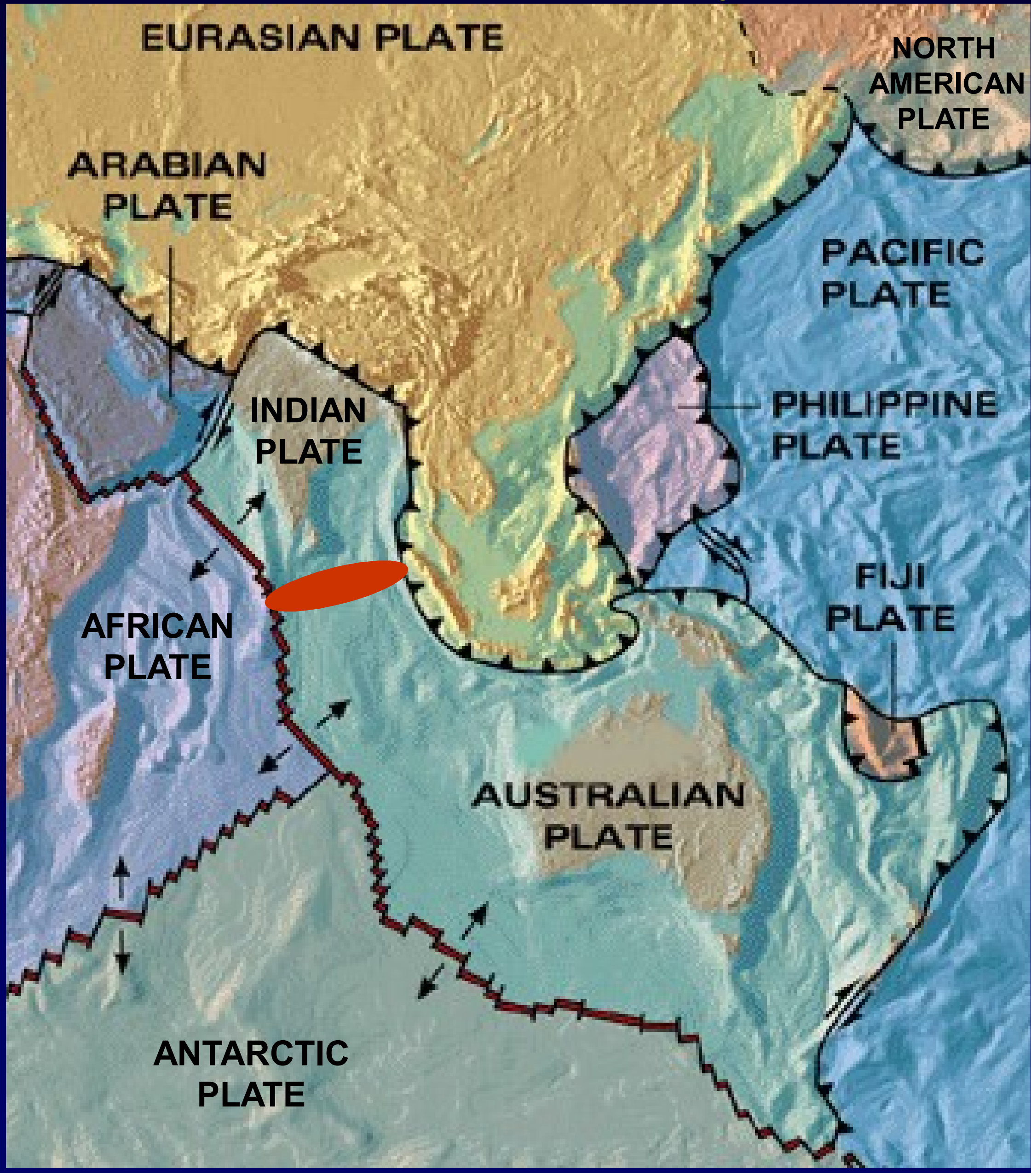


Plate-boundary Zone

- In some regions, the boundaries are not well defined.
- Deformation occurs due to movement of plates.
- The zone extends over a broad belt (called a *plate-boundary zone*).

Plate-boundary Zone



Earthquakes Epicenter at Plate Boundaries

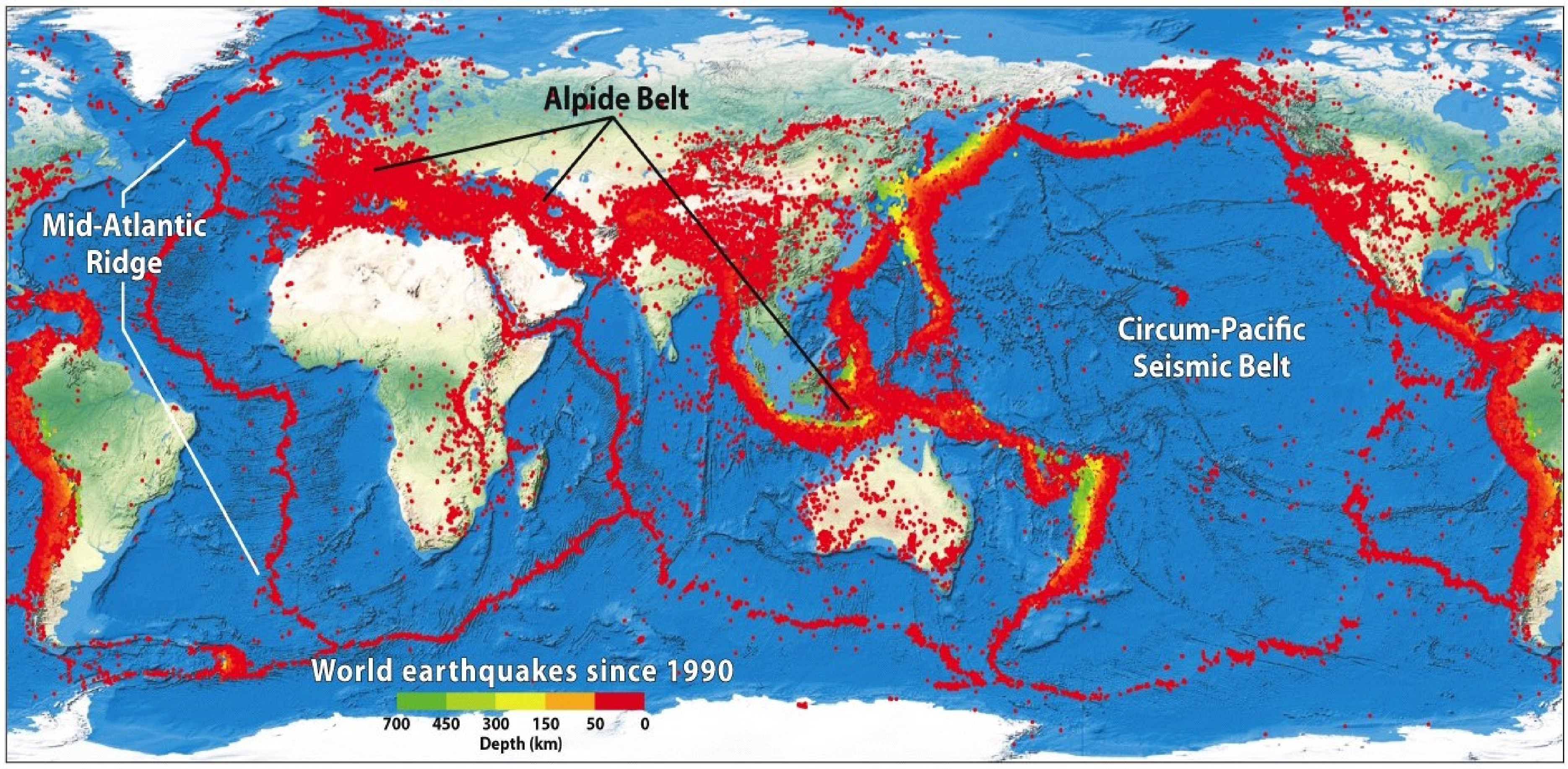


Plate Motion

Absolute

Relative

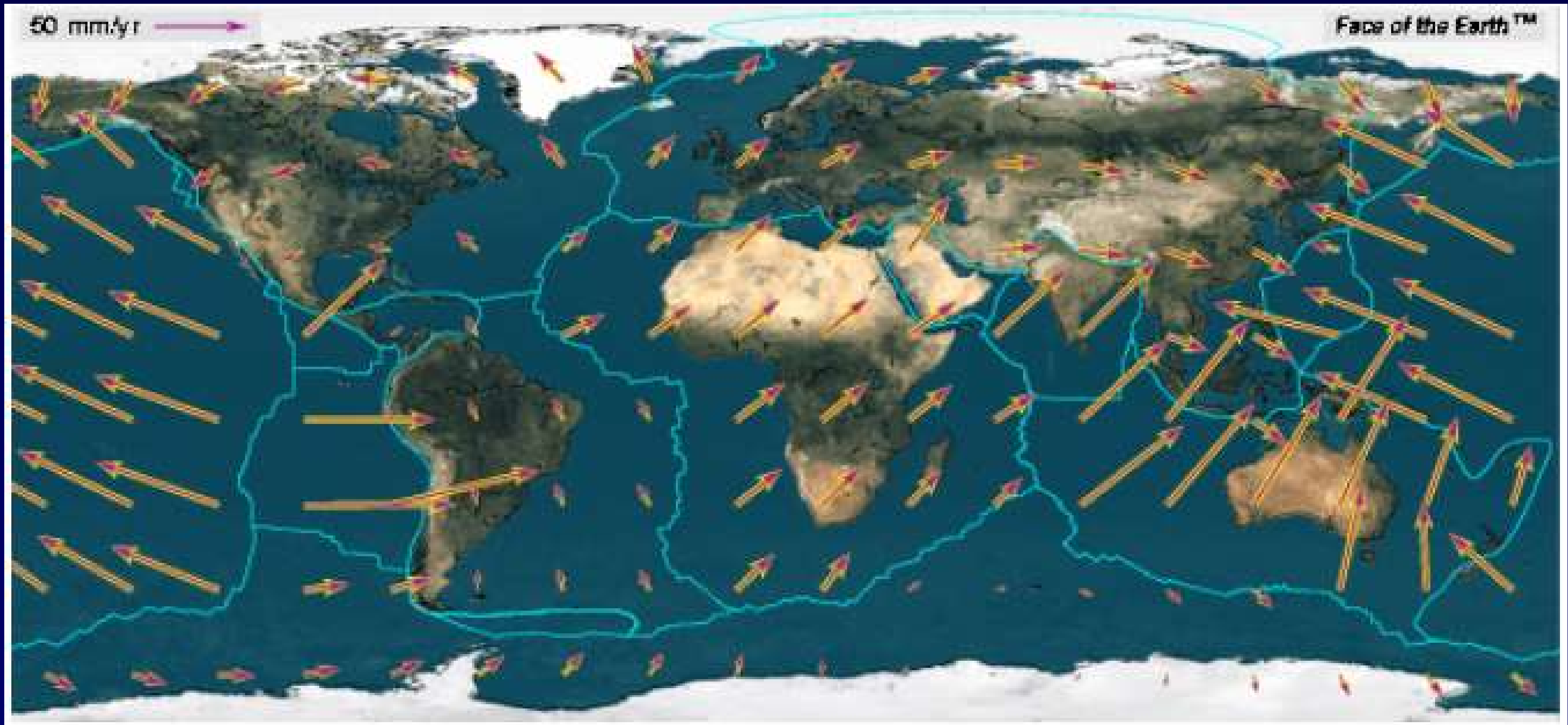
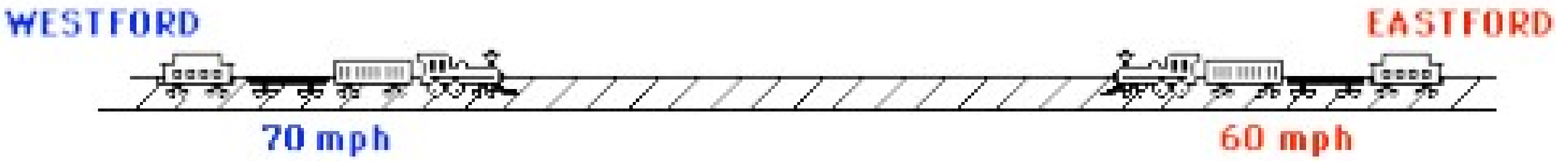


Plate Motion

relative vs. absolute motion

think about two trains traveling down the track



absolute motions:

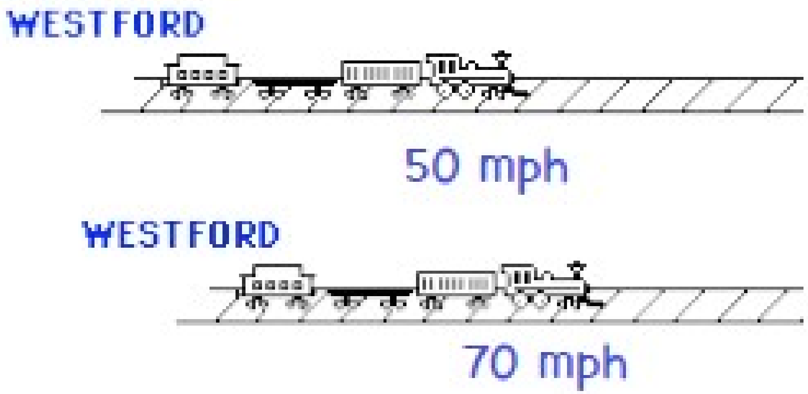
train from Westford:	70 mph to east (+ direction)
train from Eastford:	60 mph to west (- direction)

relative motion of two trains with respect to each other:
(70 mph east) - (-60 mph west) = 130 mph toward each other

Plate Motion

relative vs. absolute motion

same two trains traveling down the track from Westford



absolute motions:

train 1 from Westford: 50 mph to east (+ direction)
train 2 from Westford: 70 mph to east (+ direction)

relative motion of two trains with respect to each other:

(70 mph east) - (50 mph east) = 20 mph

Estimation of Plate Motion

- Marine magnetic anomalies (stripes)

  (From age of seafloor and distance from ridge)

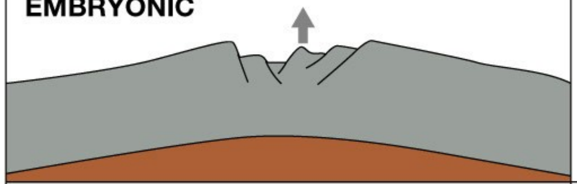
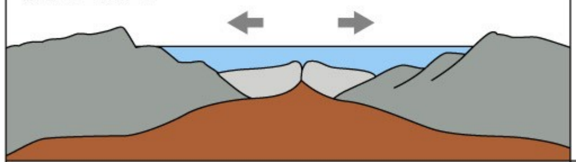
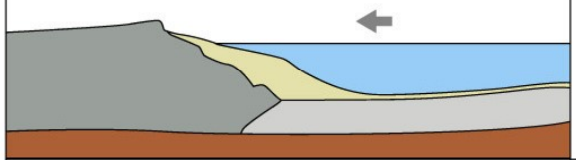
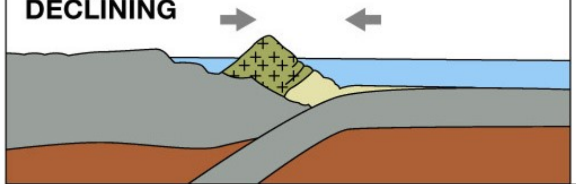
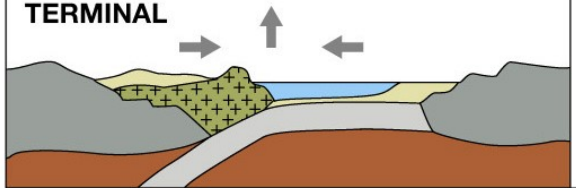
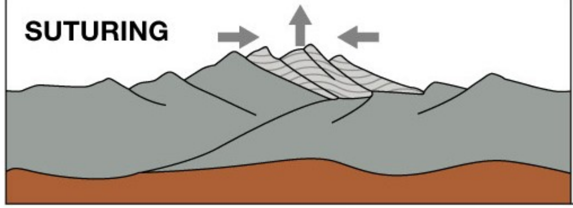
- Global Positioning System (GPS)

  (Measure location of specific point at different time)

- Hot spot tracks

  (Determine distance, age and change of orientation of hot spot track)

Life cycle of Plates →"The Wilson Cycle"

STAGE	MOTION	PHYSIOGRAPHY	EXAMPLE
EMBRYONIC 	Uplift	Complex system of linear rift valleys on continent	East African rift valleys
JUVENILE 	Divergence (spreading)	Narrow seas with matching coasts	Red Sea
MATURE 	Divergence (spreading)	Ocean basin with continental margins	Atlantic and Arctic Oceans
DECLINING 	Convergence (subduction)	Island arcs and trenches around basin edge	Pacific Ocean
TERMINAL 	Convergence (collision) and uplift	Narrow, irregular seas with young mountains	Mediterranean Sea
SUTURING 	Convergence and uplift	Young to mature mountain belts	Himalaya Mountains

Summary

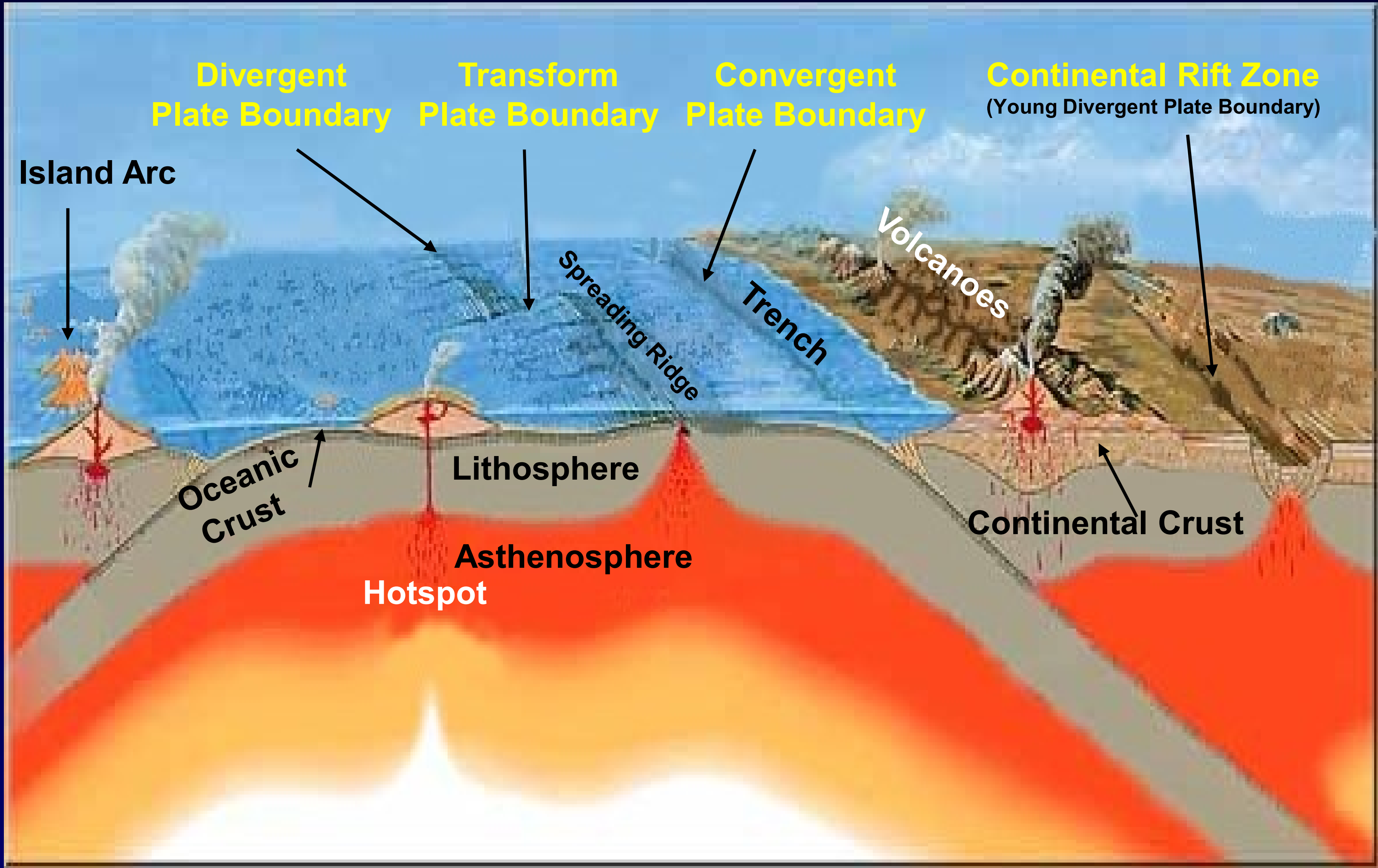
- The continents are part of tectonic plates which drift slowly at a speed of few centimeters per year over Asthenosphere.
- Asthenosphere is highly viscous, easily deformable layer between upper and lower Mantle.
- Most volcanic and earthquake activities occurs at or near plate boundaries.
- The place where the two plates meet is called a plate boundary.
- Plate interiors are relatively quiet.

- Alternate stripes of positive and negative magnetic anomalies are caused by seafloor spreading and reversals of the Earth's magnetic field.
- Symmetric & asymmetric spreading both are common phenomenon
- Spreading ridges and magnetic lineation are offset along transform faults and their fossil traces (fracture zones).
- Oblique Pseudofaults.
- Spreading ridges and magnetic lineation are also obliquely offset along Pseudofaults.

- Micro-continents.
- Details of structure of continental and oceanic domains
- Better understanding of how plate move.
- And many more……………….!

Do you know ?

- Oldest ocean floor → ~200 million years old
- Land → ~4.54 billions years old
- Why is seafloor so young relative to continents ?
- Is the Earth expanding due to seafloor spreading ?



We should live in harmony with nature

Pillow lava at CIR observed during manned submersible dive